



Fake review detection on online E-commerce platforms: a systematic literature review

Himangshu Paul¹ · Alexander Nikolaev¹

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Abstract

The increasing popularity of online review systems motivates malevolent intent in competing sellers and service providers to manipulate consumers by fabricating product/service reviews. Immoral actors use Sybil accounts, bot farms, and purchase authentic accounts to promote products and vilify competitors. Facing the continuous advancement of review spamming techniques, the research community should step back, assess the approaches explored to date to combat fake reviews, and regroup to define new ones. This paper reviews the literature on Fake Review Detection (FRD) on online platforms. It covers both basic research and commercial solutions, and discusses the reasons behind the limited level of success that the current approaches and regulations have had in preventing damage due to deceptive reviews.

Keywords Cyber-security · Opinion spamming · Fraud · Spam detection · False information · Disinformation

1 Introduction

User-generated content already influences many consumers' purchasing decisions, helps sellers maintain high reputation and profits, informs vendors about the activity of their peers and helps them swiftly react to any service quality issues (Jindal and Liu 2008). Spiegel Research Center reports that over 90% of consumers refer to online reviews before making purchases.¹

¹ How Online Reviews Influence Sales. <https://spiegel.medill.northwestern.edu/online-reviews/index.html>.

Responsible editor: Johannes Fürnkranz

✉ Himangshu Paul
himangsh@buffalo.edu

Alexander Nikolaev
anikolae@buffalo.edu

¹ Department of Industrial & Systems Engineering, University at Buffalo, New York, USA

To counter the positives, the capability of propagating information to the masses in real-time at minimum cost also opens the door to fraudsters for spreading false information, and in particular, fake reviews. Fake reviews can be classified as opinion-based disinformation that is fabricated and disseminated by malignant actors—review spammers—with the intent to mislead and deceive consumers. Here, note first that, while fact-based false information (e.g., false news, or hoaxes) can be fact-checked, and hence, confirmed or disproved, fake reviews cannot: indeed, in posted reviews, people report their unique, individual experiences, and express *opinions* and *preferences* as opposed to verifiable facts. Second, fake reviews fall into the category of “disinformation”—false information created to intentionally deceive its consumers,—as opposed to “misinformation”, which is false information created without the intent to mislead (Kumar and Shah 2018). Third, note that not all paid reviews are automatically fake. For example, even highly reputable online vendors incentivize their customers to submit reviews for recently purchased products; such reviews may be marked as “collected as part of a promotion”. What matters for the classification of reviews is the amount of pay offered, the number of paid reviews solicited or allowed to be submitted, and importantly, whether consumers are offered such incentives just for submitting a review, or for submitting a necessarily positive review.

Honest vendors and product providers are obvious stakeholders in the fight against fake reviews, as different types of actors look to exploit publicly accessible product/service review venues to promote goods, brands, and online business accounts (Heydari et al. 2015). Fake reviews deceive consumers into unfounded beliefs about product quality, which leads to customer dissatisfaction (Lim et al. 2010). Moreover, in a type of cyber-warfare, sellers use online review systems as a tool to attack and vilify competitors (Crawford et al. 2015; Luca and Zervas 2016). This “unfair play” can be rather intricate: e.g., some Amazon sellers have been suspected to get their rivals suspended by buying fake five-star reviews for those rivals’ products.² Two facts about review-based service Yelp.com speak to the motivation and consequences of fake review activity: (1) a one-star increase in Yelp rating corresponds to a 5–9% increase in revenue (Luca 2012); and (2) roughly 16% of restaurant reviews on Yelp are filtered out as fake (Luca and Zervas 2016).

Besides disrupting decision-making of individual consumers and threatening the welfare of honest vendors (Ott et al. 2013), spread of fake reviews affects public trust regarding the entire online platform (Vidanagama et al. 2020). Hence, mitigating the impact of malignant actors responsible for fake content has become an important task for review platforms as business units, in their strife to preserve the interests of honest users and sellers, and ultimately, own market value.

Fake Review Detection (FRD) research focuses on approaches toward making e-commerce review platforms well protected so that keeping up the spread of fake reviews would not be worth the returns (Jindal and Liu 2008; Lim et al. 2010; Mukherjee et al. 2012; Ott et al. 2011). Companies that own Online Social Networks (OSNs) have also relied on their internal resources to develop methods for identifying and banning the users responsible for spreading false information: see, e.g., the reported

² Dirty dealing in the \$175 billion Amazon Marketplace. <https://www.theverge.com/2018/12/19/18140799/amazon-marketplace-scams-seller-court-appeal-reinstatement>.

case studies by Facebook (Beutel et al. 2013; Stein et al. 2011), Tuenti (Cao et al. 2012), Flipkart (Hooi et al. 2016; Kumar et al. 2017, 2018b), Yelp (Mukherjee et al. 2013c) and Renren (Yang et al. 2011; Wang et al. 2013).

Despite years of effort, we have not been able to eliminate the ill-effects of deception in review writing. This is the case in large part because no “ground truth” is there when it comes to distinguishing honest opinions from dishonest opinions. Even human evaluators are found to consistently fail to distinguish fake reviews from authentic ones (Crawford et al. 2015). In order to judge if a particular review is deceptive, it is not sufficient for an evaluator to look at this review only, i.e., in separation from other reviews of the same product, which makes review analysis a non-trivial and time-consuming task (Lim et al. 2010). Continuous advancement of spamming techniques, involvement of freelancer workforce in creating fake reviews (Fayazi et al. 2015; Yao et al. 2017), and collaboration of spammers in groups (Mukherjee et al. 2011, 2012; Xu et al. 2013; Choo et al. 2015) make FRD particularly difficult. This calls for a comprehensive analysis of the advances in the regulations and state-of-the-art techniques that have been employed to curb the influence of review spamming.

The objective of this paper is to chronicle the evolution of the FRD state-of-the-art focusing on the conceptualization of approaches in reference to footprints/cues (observable *features* of review content and context that can support inference) and user behavior-based modeling assumptions. With the mind to assist the audience that seeks a transparent yet comprehensive review, this systematic literature review explains the logic behind the FRD modeling approaches developed for online review-based e-commerce platforms, as they progressively increased in conceptual complexity in response to the evolving spamming techniques. While prior literature reviews have focused on the technical aspects of FRD modeling, this paper emphasizes the developing shift in perception about how spammers work to imitate honest reviewers when crafting “believable” fake reviews. The contributions of this paper can be summarized as follows.

First, this paper provides a systematic review of the literature on FRD. We reach into the bodies of literature in such fields as information systems, e-commerce, cybersecurity, and machine learning. Viewing and explaining spamming activities from different angles, we perform a categorization by modeling approaches. We also categorize the adopted technical approaches—e.g., those relying on application-specific models, social network considerations, and pattern recognition,—by training mode, predictive features considered, and algorithms employed. Further, we review the current practices and solutions employed for user feedback collection and aggregation by the active consumer product platforms.

Second, this paper stresses that the existing methods and regulations have not yet been able to eradicate the damaging effects of fake review activity in practice. In doing so, we point at the difficulties associated with combating the different types of malignant influencers.

Third, in conclusion of this systematic literature review, we indicate the necessity of exploring other, alternative approaches to reduce the damage due to fake reviews. To this end, we highlight the value of decentralized user-empowered information systems where users could be presented with synthesized reviews from trusted sources only. Then, the impact of fake reviews could be mitigated without their complete eradication.

The paper is organized as follows. Section 2 lays out the terms and domain landscape of FRD. Sections 3 through 8 review FRD modeling approaches and computational techniques, and highlight their merits, benefits, and challenges. Section 9 reviews other, approach-independent but domain-specific, FRD tools. Section 10 focuses on existing commercial FRD implementations. Section 11 presents concluding remarks and reviews future research considerations, which might turn out useful for FRD in the near future.

2 FRD: domain landscape, terms and review structure

FRD is a sub-domain of opinion spam detection. Opinion spam detection also includes spreading rumors/hoaxes for indirect financial gain, spreading propaganda news for political gain, etc., where FRD deals with deception on online review platforms, i.e., false information in reviews posted online. It includes both *spammer* detection and *spam* (or *spamming*) detection problem types.

In regards to commonly adopted FRD *terms*, we observe the following. First, fake reviews have been referred to as opinion spam, deceptive/ untruthful/ fraudulent/ fictitious/ false/ shill/ spam reviews and “manipulated offerings”—among other, much less frequently used terms. Similarly, a “bad actor” involved in writing and spreading fake reviews is referred to as review/opinion spammer, opinion fraud, fraudster, fraudulent user, fake reviewer, deceptive reviewer, malicious user/account, etc. For consistency, we will use the terms (review) spammer, fake review, and spamming throughout this paper to refer to a “bad actor”, their deceptive reviews and their malicious activity, respectively. Second, a review that is not fake is typically referred to as authentic, unbiased or truthful. Reviewers that post such reviews have been referred to as truthful, trusted, genuine or authentic; of these, this paper will consistently use the terms “truthful review” and “truthful reviewer”. Third, other relevant FRD terms include “singleton” (aka. *one-time*) review spamming (where a spammer creates a fake account on a review platform to leave exactly one fake review and then never use this account), “sock-puppet” accounts (where a spammer creates and manages multiple fake user accounts to target the same product/business), and “group of spammers” (where multiple spammers collaborate to amplify the influence on the same target product(s)/business(es)).

In this Systematic Literature Review (SLR), we pay close attention to both FRD *modeling approaches* and *technical methods*. Step 1 of the SLR was a “Google Scholar³” search to understand the FRD domain landscape, relying on the literature reviews on FRD published within the past five years (Wu et al. 2020; Ren and Ji 2019; Hussain et al. 2019; Vidanagama et al. 2020; Kumar and Shah 2018; Dewang and Singh 2018; Crawford et al. 2015; Heydari et al. 2015). Expanding the discussion of these already existing surveys, note that they vary in scope: some go beyond FRD only (Wu et al. 2020; Kumar and Shah 2018), while some focus on specific FRD techniques: e.g., machine learning techniques (Crawford et al. 2015; Dewang and Singh 2018; Hussain et al. 2019), neural networks (Ren and Ji 2019), etc. Also, few of the

³ Google Scholar <https://scholar.google.com/>.

prior literature reviews are systematic: only Heydari et al. (2015), Hussain et al. (2019), and Wu et al. (2020) also did SLR. A typical SLR is exhaustive and transparent, in that it is based on specific disclosed search-keywords and inclusion/exclusion criteria, and also, is reproducible, which altogether helps avoid any unexplainable omissions. Compared to its counterparts, our work is of value due to several considerations: (1) it captures the seminal contributions to FRD and classifies all influential FRD papers as well, in a way that is comprehensive and yet non-technical, which must be welcomed by newcomers into this mature field; (2) it focuses on the evolution of FRD techniques, including every important technical approach, while avoiding algorithmic details; (3) it explicitly cross-matches the FRD modeling and technical approaches; (4) it includes recent FRD techniques such as neural network-based approaches, which were not introduced until 2015; and (5) it distinguishes the *fake-review*-focused FRD literature from the *fake-reviewer*-focused literature.

Step 2 of our SLR was to identify a set of (synonym) terms and keywords adopted in the FRD domain, and to use these keywords to ignite the systematic search for manuscripts with two search engines, “Web of Science” and “Engineering Village”. We sought for manuscripts that were published in English between 2007 (the seminal FRD work was published in 2007) and 2019, and whose titles contained at least one combination defined by the following search string: ((‘Unusual’ OR ‘Manipulat*’ OR ‘Spam*’ OR ‘Fake’ OR ‘False’ OR ‘Fictitious’ OR ‘Fraud’ OR ‘Decepti*’ OR ‘Shill’ OR ‘Suspicious’) AND (‘Review*’ OR ‘Opinion*’ OR ‘Account*’) AND (‘Detect*’ OR ‘Find*’ OR ‘Uncover*’ OR ‘Pattern*’ OR ‘Spot*’ OR ‘Discover*’ OR ‘Footprint*’ OR ‘Identif*’ OR ‘Filter*’)). The Step 2 search returned 482 “Web of Science” and 278 “Engineering Village” manuscripts. After scanning their abstracts, and upon dismissing duplicate and irrelevant records, we ended up with 109 manuscripts.

In Step 3, qualitative evaluation of the 109 manuscripts was performed by scanning their introductions, locating contributions, and ruling out unimpactful (duplicate in contribution) manuscripts, considering collectively the modeling and methodological elements, implementation/policy elements and reported benchmarking. As a result, 20 records were rejected while 89 manuscripts were selected into the seed set for a subsequent snowball search.

In Step 4, we performed a backward snowball tracking from the literature in the seed set. This revealed 10 additional relevant manuscripts, making it 99 total as the primary set for detailed analysis. Figure 1 summarizes our material selection process in a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram, following Moher et al. (2009).

In Steps 2-3, we focused only on the state-of-the-art fake review/reviewer detection/filtering works in application to consumer-review-based and/or e-commerce platforms, excluding other aspects of opinion spamming, fake account detection, and spamming in OSNs in general. Examples of the excluded topics are: online deception in social media (an umbrella term for financial fraud, insurance fraud, health service fraud, etc.), fake news detection, spamming in online discussion communities/forums, opinion mining and sentiment analysis. Note that some topically relevant articles did not make it into the final set due to our judgment call: e.g., we bypassed some purely technical papers, and papers of low impact on the FRD field. We used the following guidelines to inform our judgment in such exclusions: (1) perceived merit/value of the

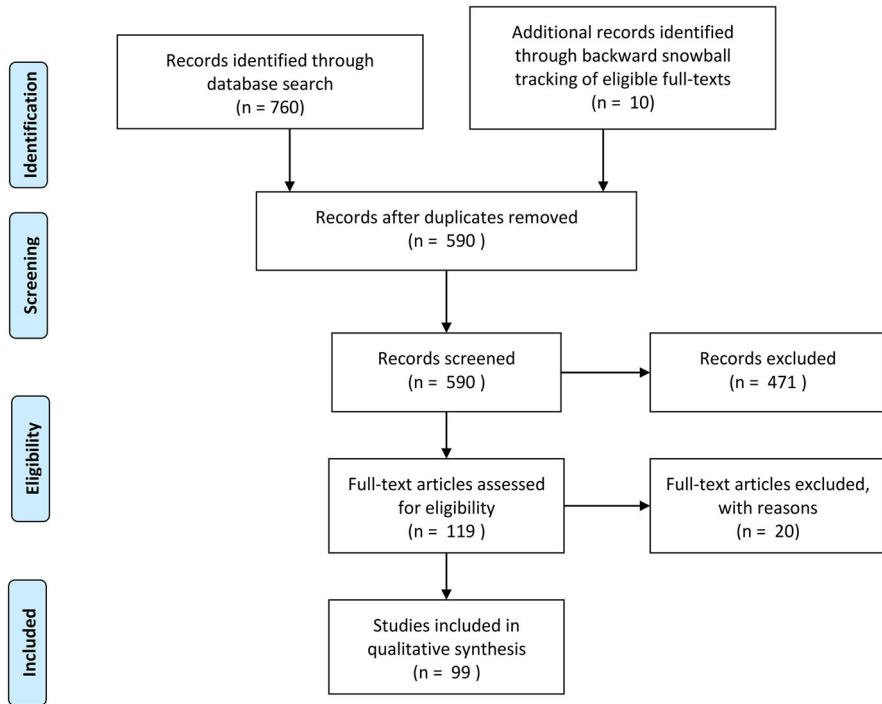


Fig. 1 PRISMA Flow Diagram of the material selection performed

work based on the conceptual/modeling and/or technically fundamental contributions; (2) number of citations (minding the year of publication and outlet quality); and (3) type of results reported (for the papers with a technical focus). We believe that the collection of articles we have selected for review gives a comprehensive representation of the important points in the evolution of FRD literature, and thus, meets the objective of this SLR.

We categorize FRD *modeling approaches* by the researchers' assumptions regarding how fake reviews are created and how their creators behave. We organize approaches into six streams: FRD models exploiting (1) content similarity (FRD-CS); (2) writing footprints (FRD-WF); (3) behavioral footprints (FRD-BF); (4) network footprints (FRD-NF); (5) rating footprints (FRD-RF); and (6) collusion footprints (FRD-CF). All these approaches share the same fundamental assumption: spammers behave distinctly differently from authentic users, and hence, it may be possible to identify fake reviews along with the spammers from the "footprints" they leave. Footprints designate textual/writing, behavioral, temporal, rating- and relationship-based clues/patterns. In trying to detect such footprints and define features to track them, researchers made numerous hypotheses, observations, and then, modeling assumptions regarding the nature of spamming as intentional activity. Some of these are general, others are business-specific.

In the upcoming sections, we present the modeling approaches in an approximate chronological order. We explain the general perception of the advocates of each approach towards how spammers operate, and discuss how this perception is translated into modeling assumptions. Where possible, we also highlight the key merits and disadvantages of each approach.

The *technical methods*—data analysis techniques and algorithms—that enable or facilitate putting models to use—are reviewed in the subsections under each modeling approach, with the focus on classification and pattern mining (aka. anomaly detection). Classification algorithms are further categorized by training mode (e.g., supervised and semi-supervised), and pattern mining algorithms—by feature type (e.g., textual, behavioral, distributional, temporal). Since the same techniques are often employed in multiple modeling approaches, we discuss each technical method under the modeling approach for which its use has been most influential. In some works, additional tools are employed, either to overcome the existing shortcomings associated with a technical approach or to complement it: e.g., dataset development is required for supervised classifiers, inference techniques are essential for probabilistic graphical models, etc. Such points are discussed in a separate section called *Implementation-Related Tools*. Some of the papers reviewed report performance comparisons of FRD algorithms. Respectively, note that the common criteria/measures adopted for FRD algorithm testing are precision, accuracy, recall, F-measure and Area Under the Curve (AUC); supervised FRD classifier comparisons mainly rely on the latter (Hussain et al. 2019). The reader may refer to Hussain et al. (2019), Dewang and Singh (2018), and Vidanagama et al. (2020) for a more detailed coverage of specific measures used in individual FRD articles.

We summarize the contributions of all the reviewed FRD articles in the tables in the “Appendix” by modeling approach, technical approach, implementation-related tool(s) used, and also, by target for detection: fake reviews vs. spammers vs. groups of spammers. Appearing at the beginning of Sect. 11, Fig. 10 visualizes the chronology of the FRD research in years 2007 to 2019. It captures the key milestones along with a list of most influential FRD articles, referencing the approach that each such article advanced and its citation count as of early 2021.

3 Models exploiting content similarity (FRD-CS)

Early FRD efforts assumed that fake reviews would be too difficult to separate from authentic ones by examining how they express opinions, and hence, focused on identifying *non-reviews* and *duplicate* reviews. A simple approach is to view fake reviews as messages that are repetitive, i.e., (near-)duplicate, or lacking any word content (Jindal and Liu 2008; Lai et al. 2010b). The latter are termed *non-reviews* or irrelevant reviews; see examples in Figs. 2 and 3.

The first approach, adopted in 2007 by Jindal and Liu (2007a, b, 2008), suggests that non-reviews must show a high degree of pairwise similarity, which may allow for spam detection. Similarity can be assessed via using a distance metric—a function of some characteristics of a posted review, e.g., its textual features, or feature counts. More specifically, in line with this approach, Jindal and Liu (2008) assume that non-

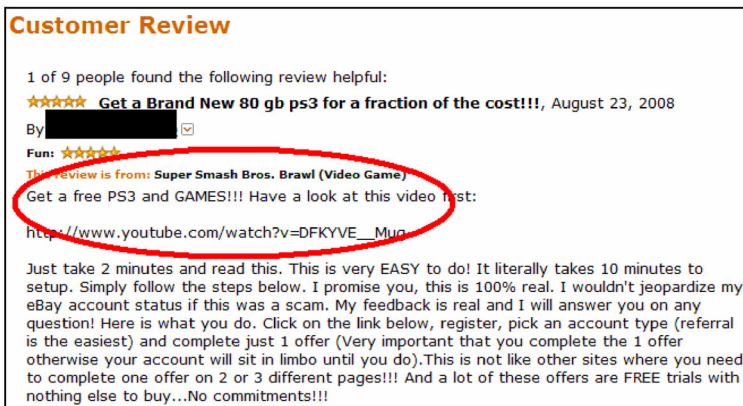


Fig. 2 An example of a longer non-review (Lai et al. 2010a)

Type	Example of Non-reviews
Advertisement for the product	"Detailed Product Specs: Standards * 802.11g, 802.11b, INMPR Compliant, TCP/IP, UPnP AV 1.0, USB 2.0,"
Advertisement for a different seller	"This is a great product but can be bought for less at: compuplus.com"
Question about the product	"What port it is for? AGP or PCI Express?? From the looks of the picture it seems like the PCI Express version. Can anyone confirm this?"
Comment about other reviews	"This Other Review is too funny."
Unrelated random text	"Go Eagles Go."

Fig. 3 Examples of shorter non-reviews (Jindal and Liu 2007a)

reviews show similarity in amount and relevance of feedback delivered, in length, in sentiment (as proportion of positive versus negative opinion bearing words), and also, that they exhibit a tendency for excessive use of numerals and capitals (overall indicating poor writing). Algur et al. (2010) assume that spammers use conceptually similar words to describe product features. Conceptual-level similarity measures are then employed to reveal such duplicate reviews. Lau et al. (2011) and Lai et al. (2010b) hypothesize that spammers leave describable language clues while writing non-reviews, and that these clues can be identified and leveraged. Consequently, they use syntactical features (e.g., *percentage of nouns, verbs, pronouns, etc.*), lexical features (e.g., *sentiment orientation, lexical diversity, lexical validity, etc.*), and stylistic features (e.g., *capitalized diversity, repetition diversity, passive voice frequency, etc.*) to detect non-reviews.

Another consideration that is also accounted for in FRD-CS is objectivity of information. The assumption here is that reviews that offer too much objective information, as opposed to being subjective, are likely to be commercial advertisements, which provides another detectable non-review clue (Li et al. 2011). See Fig. 3 for examples of advertisement-type non-reviews consisting of only objective information (e.g., product specification).

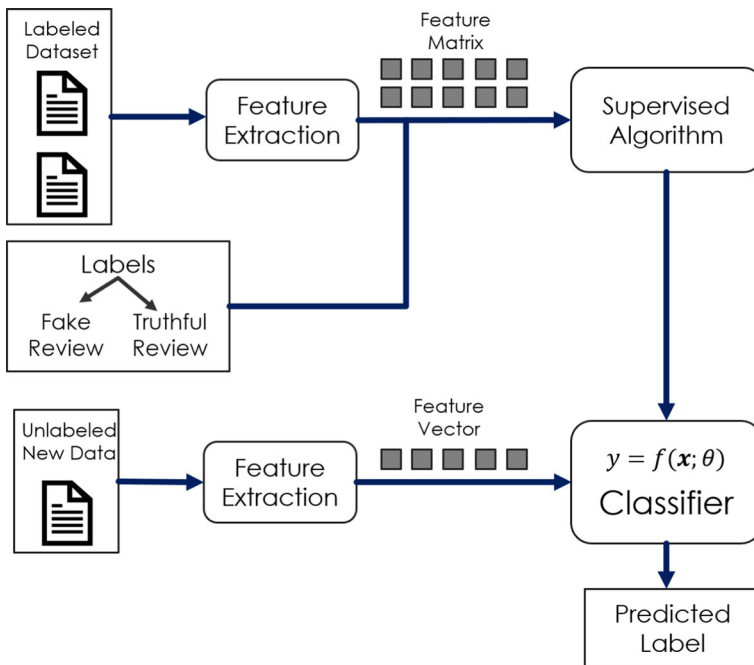


Fig. 4 A conventional Supervised Classification framework

3.1 Supervised learning techniques

FRD can be seen as a classification problem with two categories, *spam* vs. *non-spam* (Jindal and Liu 2008), or *fake* vs. *genuine* (Wahyuni and Djunaidy 2016). Ott et al. (2011) approach FRD from three perspectives: (1) as a standard text categorization task; (2) as a psycho-linguistic deception detection task; and (3) as a problem of genre identification, where deceptive and truthful writing can be viewed as sub-genres of imaginative and informative writing. Figure 4 schematically depicts the logic of supervised classification for FRD.

Cardoso et al. (2018) conduct a comparative analysis of content-based classification methods for FRD. They consider two strategies: offline learning and online learning. They find that over time, FRD algorithms degrade in effectiveness in practice (unless frequently updated), and thus, claim that online learning methods are to be favored as more scalable, and also, adaptable to be able to respond to continuously evolving spamming techniques.

Supervised and semi-supervised *machine learning* models can be used either to classify reviews as fake vs. genuine, or to estimate the likelihood of a review to be spam (Jindal and Liu 2008). Labeled training data (real-world examples) of both fake and genuine reviews are required to build such classification models. Examples of supervised classifiers widely-used for FRD include Logistic Regression, Naïve Bayes, Decision Trees, Random Forest, Boosting classifier, and Support Vector Machine, among others.

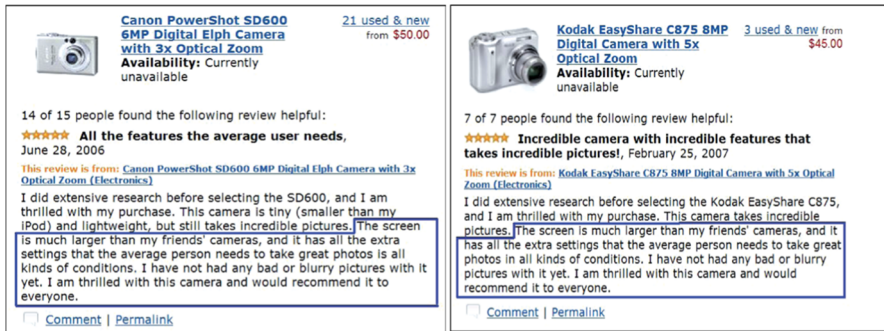


Fig. 5 Example fake reviews with highly overlapping content (Lau et al. 2011)

Supervised classifiers automatically identify fake reviews and are most widely-used tools for FRD. However, gathering ground truth training data is a major challenge for building an effective supervised classifier. Typically, human annotators or rule-based heuristics are used for this purpose (Fayazi et al. 2015), so biases and/or limiting assumptions are hard to avoid: see Sect. 9. Another issue with supervised learning concerns the identification of informative features, which are not immediately obvious in fake reviews (Lai et al. 2010b). Also, as elaborated in Sect. 9, hand-picked discrete features often struggle to capture complex semantic information (Ren and Zhang 2016).

3.2 Textual pattern mining techniques

Probabilistic Language Modeling: Some works under the FRD-CS modeling approach use language modeling for automatic detection of fake reviews. The underlying assumption here is that the degree of “untruthfulness” in a review can be estimated based on the “overlap” in the semantic contents of reviews. Figure 5 shows two such (similar) fake reviews posted for different products. If the content of one review appears to be similar to that of another, the likelihood of both reviews being untruthful is assumed to be high (Lai et al. 2010b; Lau et al. 2011). In other words, “overlap” of review contents can be used to inform classification features to identify fake reviews.

Lau et al. (2011) apply an unsupervised text mining model and a semantic language model together to detect review content overlapping. The text mining model is mindful of high-order concept association relationships (e.g., “fabulous” → “fantastic”), and the language model accounts for semantic term relationships (e.g., “love” → “like”). Negative Kullback–Leibler Divergence is used to quantify the similarity. Taken together, these models comprise a method that measures the “edit distance” between different wordings and can recognize even two “seemingly different” reviews, e.g., by the length, as near-duplicate.

Note, however, that “high similarity” in content with other reviews does not necessarily make a review “fake”, and assuming so might cause some truthful reviews to be flagged as “fake”. Sometimes, truthful reviewers copy their own reviews when providing feedback, e.g., for similar products or for different versions of the same

product. For instance, a truthful reviewer may write a review for a book and post the same review both for the paper and e-book versions of this book.

4 Models exploiting writing footprints (FRD-WF)

Models in the next emerging wave share the assumption that spammers are unable to write reviews with the same degree of expressed emotion as honest reviewers, which gets reflected in the text they produce. The conjecture that reflects this position is that authentic users and spammers use the same words at different frequencies. Extending this logic, according to Xue et al. (2015), an authentic review is one that is of “right” length and level of detail, is “balanced” and consistent in its essence, which altogether makes it helpful. These and various other writing footprints of spammers are identified and leveraged under this FRD approach.

The first work based on FRD-WF was presented by Ott et al. (2011) with a view to identify fake reviews that have been deliberately written to sound authentic. Prominent works under FRD-WF define and rely on advanced textual features to categorize deceptive and truthful opinions, including *word n-gram* features (Li et al. 2011, 2014b; Ott et al. 2011, 2013; Feng et al. 2012a; Shojaei et al. 2014; Najada and Zhu 2014; Hai et al. 2016; Wang et al. 2016a), *character n-gram* features (Fusilier et al. 2015b; Mukherjee et al. 2016; Dong et al. 2018), *parts-of-speech* features (Feng et al. 2012b; Banerjee and Chua 2014b, a; Banerjee et al. 2015; Rout et al. 2017b), other *linguistic* features (e.g., *average number of words per sentence*, *number of paragraphs*, *rate of misspelling and swearing*, etc.) (Ott et al. 2011; Harris 2012; Shojaei et al. 2014; Banerjee and Chua 2017; Plotkina et al. 2020), *deep linguistic* features or *sentence-structure-based* features (e.g., *parts-of-speech tag bigrams*, *dependency parsing features*, *context free grammar parsing features*, etc.) (Xu and Zhao 2012; Feng et al. 2012a; Mukherjee et al. 2013b), *aspect-based* features (Feng and Hirst 2013) and groups of *psychological* features (Ott et al. 2011; Plotkina et al. 2020), among others.

A common belief suggests that writing fake reviews is more challenging than writing authentic ones, due to spammers’ personal disinterestedness as well as self-imposed limitation on the time taken to write reviews. For this reason, spammers can be assumed to more often use short simple sentences and fewer average syllables per word, and produce short and easily comprehensible reviews (see Fig. 6). On the other hand, in looking to avoid being detected through these traits, skilled spammers can produce too complex, overly polished reviews. Per this logic, the average number of words per sentence can be viewed as a feature to help distinguish authentic text from unauthentic.

Misspelling deserves a special consideration. From one perspective, a hired spammer can be more professional; from the other, they can be asked to make mistakes on purpose, to make the writing appear less formal. Then again, the latter intent can be overly exaggerated. To identify such traits that may have spam prediction power, researchers measure *readability/comprehensibility* (measure of effort required to understand a review) and use features such as *number of characters per word*, *number of words per sentence*, and *number of long words* (Harris 2012; Banerjee and Chua 2014b, a; Banerjee et al. 2015; Banerjee and Chua 2017). To give specific examples, Harris (2012) analyze three linguistic qualities—*readability*, *sentiment*, and

Suggestion tebavor Version 1 - Sep 13, 2011 My face... great, but what I really want to know about are my breasts!	★★★★★
Satisfied tebavor Version 1 - Sep 17, 2011 I'm extremely satisfied with my caricature. Well done.	★★★★★
So great! tebavor Version 1 - Sep 17, 2011 Seems a real one. It's a lot of fun	★★★★★
Good app tebavor Version 1.0 - Sep 17, 2011 Not as stupid as I thought!	★★★★★
Really happy tebavor Version 1.0 - Sep 17, 2011 It's actually a personal one on one reading. I'm really happy with the outcome.	★★★★★
Ooh la la Muquiwara78 Version 1 - Aug 25, 2011 Qd I see the head of my old guy, it scares me. At the same time, I will be like!	★★★★★
Satisfied Muquiwara78 Version 1 - Aug 25, 2011 I'm extremely satisfied with my caricature. Well done.	★★★★★
ha ha ha Muquiwara78 Version 1 - Aug 25, 2011 I want to be that close of J Lo!	★★★★★
Simple IQ test Muquiwara78 Version 1.0 - Aug 25, 2011 Good app.	★★★★★
Thanks Muquiwara78 Version 1.0 - Aug 25, 2011 I'm extremely satisfied	★★★★★
Rides merquezcito Version 1 - Sep 1, 2011 I just discovered old	★★★★★
A beautiful gift merquezcito Version 1 - Sep 1, 2011 I got one made as a gift for my god-daughter	★★★★★
Perfect merquezcito Version 1 - Sep 1, 2011 Nothing else to say	★★★★★
Good app merquezcito Version 1.0 - Sep 1, 2011 Not as stupid as I thought!	★★★★★
Deadly manul Version 1 - Oct 23, 2011 It does not care a shot of being old. Fortunately it is not immediately	★★★★★
Good job manul Version 1.1 - Dec 3, 2011 I really like this app. I want another!!!	★★★★★
ha ha ha manul Version 1 - Oct 23, 2011 I want to be that close of J Lo!	★★★★★
Simple IQ test manul Version 1.0 - Oct 23, 2011 Good app.	★★★★★
Thanks manul Version 1.0 - Oct 23, 2011 I'm extremely satisfied	★★★★★

Fig. 6 Examples of review spammers writing many short reviews within a short time period and re-using the same text (Akoglu et al. 2013)

polarity,—while Banerjee and Chua (2014b) define two metrics for measuring *readability* of a review: *complexity* and *reading difficulty*.

Spammers may generate numerous fake reviews automatically by copying sentences from truthful reviews. Sun et al. (2013) apply this technique to generate synthetic fake reviews and show that many existing feature-based FRD models fail to identify these synthetic reviews as fake. With a view to identify this kind of review manipulation, a defense mechanism framework due to Sun et al. (2013) leverages the difference of semantic flows between fake and truthful reviews. The framework is based on two types of coherence analyses: pairwise sentence coherence and multiple sentence coherence. To measure sentence coherence, such features as *sentence transition*, *word co-occurrence*, *word/semantic similarity*, *semantic dispersion*, etc., come useful.

FRD-WF works also assume that spammers often write fake reviews without having hands-on, i.e., detailed information about the product, or without real experience with the service being reviewed. Intuitively, fake review content should tend to avoid making specific points of critique, which means that fake reviews should be less diverse (Xu et al. 2015), and contain statements contradictory to the actual characteristics of the product (Feng and Hirst 2013)—all based on what contextual knowledge the spammers may have about it (Zhang et al. 2018; Liu and Pang 2018). However, a totally opposite assumption also has its merits, i.e., the one suggesting that fake reviews can be purposely set up to contain more details compared to truthful reviews. Detecting such fake reviews also requires measuring the frequency of product-related terms used in review text. Overall, any anomalous observation might raise a FRD-WF model's suspicion. Specific to this approach, researchers analyze the linguistic property *specificity/informativeness* (which refers to the level of details mentioned in reviews) to track product-specific clues characteristic of fake reviews (Ott et al. 2011; Banerjee et al. 2015; Banerjee and Chua 2017), and *lexical diversity* to track the *frequency of spatial and temporal words* in review texts, as well as *proportions of various parts-of-speech tags*. Feng and Hirst (2013) use *profile compatibility* features that express the level of compatibility of personal experience shared in a review with the collection of reference information (collective profile) about the business/product. They analyze two types of compatibility measures: *existence of distinct aspects* and *description of general aspects*.

Other FRD-WF works rely on the interplay between sentiment and deception. Here, researchers argue that, since spammers are typically hired to alter customer perception towards some product/service, then the *sentiment polarity* (i.e., positive, neutral and negative) showcased by a spammer should be different from that of the rest of the crowd. In this view, reviews that express only positive or negative sentiments, rather than mixed/neutral sentiments, are assumed more likely to be spam (Li et al. 2011). In order to make their statements more influential, spammers may tend to use *exaggerated language* (Ott et al. 2013), and *overuse capital letters and punctuation signs* (Banerjee and Chua 2017). Relevant textual features signaling of *exaggeration* are—*proportions of positive/negative cues*, *perceptual words* and *future tenses* used in reviews (Banerjee and Chua 2014a, b). Peng and Zhong (2014) focus on the importance of sentiment analysis for FRD based on three observations: (1) at times, inconsistency shows between the submitted ratings and expressed sentiment polarity in fake reviews; (2) sentiment strength expressed in a typical fake review fluctuates considerably; and

(3) sentiment strength will be different between two reviews with same rating scores that have a different number of sentences. For example, a review with lots of sentences providing detailed information will be more convincing (i.e., have a higher sentiment strength) than a review with only a couple of sentences and without any details. Dong et al. (2018) classify reviews based on extracted (from review text) topic features as well as corresponding sentiment features, assuming that reviewers' sentiment varies over topics, e.g., product categories. Liu and Pang (2018) also use *product-specific aspects* (e.g., price or size) and *expressed sentiment* to enable content-based review screening.

Another commonly adopted assumption states that false statements cannot hide the psychological clues of lying, e.g., increased negative emotion and psychological distancing. Further, differences can be identified between positive fake reviews—those that intend to promote or restore reputation,—and negative fake reviews—those that intend to defame competitors. To this end, Ott et al. (2013) identify linguistic features that may vary with the sentiment; one such feature is the frequency of use of the *first-person-singular pronoun* in review text which may signal putting too much emphasis on the self (i.e., the writer). Indeed, they find that this frequency is lower in negative sentiment reviews than in positive sentiment reviews. The reasoning here is that a negative tone of reviews causes the reviewers to psychologically distance themselves from their negative statements. Banerjee and Chua (2017) define similar linguistic properties that reveal the cognitive challenges involved in writing fake reviews such as *negligence* (Banerjee et al. 2015). The relevant textual features here are: use of *self-references*, *uncertainty cues*, *specific fillers* and *tentative words*, etc. On a different note, Li et al. (2011) and Yoo and Gretzel (2009) point out that spammers tend to mention brand names, or product names, too often in their reviews, and also, that they write reviews from the “second person” perspective and hardly ever use self-reference. That is, fake reviews may speak too much about what “you should do”, rather than how “I experienced it” (Li et al. 2011). For further examples, refer, e.g., to Ott et al. (2011) and Li et al. (2014c) who engage in psycho-linguistic analysis using Linguistic Inquiry and Word Count (LIWC) software, and also, to Plotkina et al. (2020) who utilize a Coh-Metrix tool to analyze review-text based on cohesion, language and readability.

Over time, spammers develop skills to avoid leaving textual clues, so analyzing reviews using hand-picked textual features is becoming less effective day by day. In order to tackle this issue, researchers have started using topic models (Li et al. 2013; Sandulescu and Ester 2015; Xu et al. 2015; Ren et al. 2014; Mukherjee et al. 2016; Dong et al. 2018), neural networks (Wang et al. 2017; Li et al. 2017; Ren and Zhang 2016), and Generative Adversarial Networks (Aghakhani et al. 2018) in particular, to automatically identify subtle, non-interpretable textual clues in reviews. In a particular topic modeling approach, Li et al. (2013) leverage subtle differences in the topic-word distributions between deceptive and truthful reviews. Under the assumption that many spammers have limited creativity, and hence, must often turn to rephrasing in writing new texts, as well as overuse synonymous words, Sandulescu and Ester (2015) advocate for the use of semantic similarity measures: they work to capture subtler textual clues than those used in vector-based similarity measures such as cosine similarity. Their experimental studies reveal that a context-specific bag-of-opinion-phrases topic modeling method widely outperforms the standard bag-

of-words model in F-score. Similarly, Hai et al. (2016) assume that FRD problems in different domains (i.e., hotel reviews vs. restaurant reviews, etc.) should be attacked individually, and develop a generalized FRD formulation with domain-specific textual features.

The advantage of review text-based FRD models is that they can be put to use right after a review has been posted, which diminishes the negative impact of fake reviews in a timely manner (Li et al. 2014b). Other types of models, such as those leveraging behavioral footprints of spammers, on the contrary, need substantial data processing time to gather evidence to inform FRD.

4.1 Semi-supervised learning techniques

Since manual labeling of fake reviews to train supervised classifiers is a labor intensive task, researchers resort to semi-supervised learning techniques. Semi-supervised classification algorithms are useful when only small amounts of labeled data and large amounts of unlabeled data are available. We first highlight two most popular semi-supervised classifiers.

Semi-supervised Classifiers—Co-training Algorithm (CA): CA is a semi-supervised learning algorithm useful when only a small set of labeled items and a large unlabeled set are available for training. The algorithm assumes that the content of each item (review) can be analyzed using two distinct models, each informed by its own feature set, that will provide complementary information about the item. Ideally, the variables in the two models should be conditionally independent, and each model should deliver a high-accuracy prediction of the item class (“fake” or “truthful” review). Using labeled review examples, CA learns a separate classifier from each model first, and then, iteratively builds additional labeled training data using the most confident predictions of each classifier on the unlabeled data. Li et al. (2011) introduce CA for FRD. Rout et al. (2017a) provide a comparison of several semi-supervised learning approaches, including CA.

Semi-supervised Classifiers—Positive and Unlabeled Learning (PU learning): PU learning is a special case of binary classification, in which a classifier (e.g., NB, SVM, etc.) is trained to distinguish the cases representative of positive and negative classes from two sets of samples available for training: the positive set and a mixed set containing both positive and negative samples. Fusilier et al. (2013) introduce the idea of PU learning for FRD. They use the dataset due to Ott et al. (2011), and compose a positive training set containing fake reviews only (note that “positive” class designates “fake” in PU learning) and an unlabeled training set containing both truthful and fake reviews. Li et al. (2014b), however, point at practical issues with the original implementation, so in response, Fusilier et al. (2015a) modify their method to take into account the polarity of reviews. Consequently, Li et al. (2014b, a) report that PU learning outperforms supervised learning in recall and F-score. Ren et al. (2014) propose a semi-supervised model, *Mixing Population and Individual Property PU Learning*, to identify fake reviews from unlabeled data, using *Latent Dirichlet Allocation (LDA)* topic model and SVM. Wu et al. (2016) apply an algorithm based on hybrid PU learning, to model and classify types of spammers. Rout et al. (2017a) report

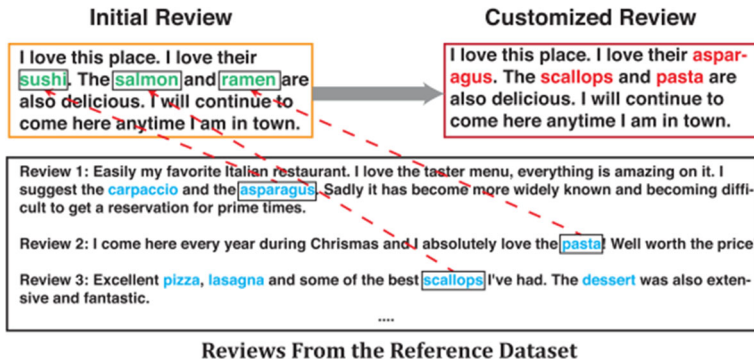


Fig. 7 Customization in automatic RNN review generation (Yao et al. 2017)

a superior performance of PU learning for FRD (based on such evaluation criteria as accuracy and F-score) in comparison with *Label Propagation and Spreading* (LPS), CA and EM algorithms.

Semi-supervised Classifiers—Other Algorithms: FRD researchers also explore other semi-supervised learning algorithms, including EM (Rout et al. 2017a), LPS (Rout et al. 2017a), Semi-supervised Multi Task Learning via Laplacian regularized Logistic Regression (SMTL-LLR) (Hai et al. 2016) and GAN (Aghakhani et al. 2018). Rout et al. (2017a) use EM algorithm to learn a supervised classification model. Hai et al. (2016) present a LogR-based multitask learning method that can boost learning for a task by sharing the knowledge contained in the training signals of other related tasks. Yao et al. (2017) show that deep learning can be used for automated generation of fake reviews. They train a Recurrent Neural Network (RNN) on a large corpus of natural language text; a restaurant review dataset is taken as a practical example. The trained RNN generates an initial fake review, which is then customized by replacing all the nouns in it with domain-specific keywords (e.g., name of a dish in a restaurant) to further capture the context for a particular entity, i.e., here, a particular restaurant (see Fig. 7).

4.2 Textual pattern mining techniques

Topic Modeling: Topic models, LDA in particular, are designed for discovering abstract topics from a collection of documents. Unlike supervised classifiers, they do not require manually pre-labeled data, although they perform best when trained on large datasets (Sandulescu and Ester 2015). Li et al. (2013) use LDA-based topic modeling approach to detect differences in the topic-word distributions between spam and truthful reviews. Xu et al. (2015) use hierarchical LDA to develop *Unified Review Spamming Model* to detect both fake reviews and reviewers in an unsupervised way. Mukherjee et al. (2016) propose a latent topic model to rank fake reviews leveraging product rating scores and timestamps of review texts. Dong et al. (2018) create a LDA-based topic-sentiment model inferring within-review distributions, over topics and sentiments simultaneously, based on the hypothesis that the topic affects the sen-

timent expressed by reviewers. They assume that reviewers express positive/negative sentiment about a service (e.g, hotel) depending on the feature-groups describing such aspects as location, cleanliness, quietness, etc. The LDA-based model extracts a topic-sentiment joint probabilistic distribution vector for each review, and then, uses this feature vector for review classification.

5 Models exploiting behavioral footprints (FRD-BF)

Looking at an entire set of reviews written by a user may reveal abnormalities in their *behavior*. Jindal et al. (2010) were first to propose an empirical model with a set of characteristic behaviors of truthful reviewers: a group of rules (short, with one to three conditions each) useful for finding unexpected reviewing patterns. Zhang et al. (2016) study verbal features of reviews and non-verbal features of reviewers—those describing the posting and social behaviors—for FRD in hotel and restaurant reviews. They report that non-verbal features have a higher predictive power: *useful votes* and *review burstiness* are the two non-verbal features found most useful. Mukherjee et al. (2013b) point at one behavioral clue found in the analysis of Yelp review data: spammers almost never submit “tips” along with a review (tips are insights about a business, written from personal experience), and hence, the number of “tips” ever given by a reviewer can be a good predictive feature. On a similar note, Liu and Pang (2018) use a set of abnormality signals from the behavioral-deviation viewpoint to identify fake reviews in Amazon review data.

In further support of the FRD-BF approach, note that at times, spammers are hired to write positive reviews so as to mitigate the impact of several truthful negative reviews, and to keep maintaining a respectable image of a product/service. Then they have to give away extreme ratings that deviate from the average rating for that product/service, or alternatively, resort to posting only one review from a large number of different accounts, i.e., posting *singleton reviews*. Thus, rating spikes and singleton review proliferation can be indicative of spamming. Wu et al. (2010a) formulate eight spam suspicion-raising criteria, and among those, identify *reactive positive singletons* (a wave of reactive positive fake reviews submitted in response to several negative reviews) and *truncated ratings* (a dip in the average product star rating observed upon removing a number of very positive consecutive reviews) as the strongest predictors of spammer activity.

Lim et al. (2010) define several behaviors characteristic of spamming, including *general-deviation* (deviation of rating provided by a reviewer from the mean rating of the product), *rating-spamming* (multiple similar ratings provided by the same reviewer on certain products), and *single-product-group-multiple-high-ratings* (multiple high ratings provided by the same reviewer to the products of the same brand within a short period of time). Fontanarava et al. (2017) propose two deviation-based features particularly useful for finding singleton reviews: *rating-variance* (standard deviation of *rating posted by a reviewer* from the *mean rating* of that product) and *timestamp-variance* (standard deviation of *timestamp of a review* from the *mean timestamp*). Li et al. (2011) find another behavioral clue: that review spammers often write fake reviews only, i.e., without hiding their activity by mixing truthful reviews with untruth-

ful, which makes their activity easier to detect. Lu et al. (2013) define and apply four binary indicators in a model that characterizes reviewer-review relations. Kumar et al. (2018a) develop an FRD model that does uni-variate and multi-variate transformations of features: this helps train nonlinear models and allows one exploit interdependence of features. Barbado et al. (2019) define four subsets of reviewer-behavioral features for analyzing Yelp reviews of consumer electronics: *social interaction* features, *personal profile*, *review activity* and *trust information* features.

FRD-BF models also consider review/feedback posting frequency as an indicator of spamming. Spammers exhibit bursts in activity: they write reviews only when being paid (see Fig. 6). Most authentic users, on the other hand, tend to contribute at a stable, albeit low, rate and take time between submitting consecutive reviews. Hence, the number of posts a reviewer submits for the same product/service/brand in a given day/week can be used as a feature for predictive modeling. Following this logic, Mukherjee et al. (2013a) introduce *maximum number of reviews* and *rating abuse* as behavioral footprints of spammers. Savage et al. (2015) consider *posting multiple reviews on the same day* as a key feature and design a binomial regression model to identify spammers. After analyzing review- and reviewer-centric features, Fontanarava et al. (2017) claim that burst features have a greater predictive power for supervised classification for FRD overall and for detecting singleton fake reviews in particular. Comparing behavioral and linguistic features, Shehnepoor et al. (2017) report that *review-behavioral* features are more useful than *reviewer-behavioral*, *review-linguistic*, and *reviewer-linguistic* ones.

Note, however, that submitting multiple ratings/reviews cannot serve as a definitive proof of spamming. It could just as well be normal for a truthful review writer to post more than one review on the same sitting, while recalling multiple purchasing experiences with the same business (Wang et al. 2012). Wang et al. (2012) also point out that behavioral features of spammers who write reviews for consumer products may be different from the spammers who write for a service/business.

5.1 Behavioral pattern mining techniques

Finding Unexpected Rules: An FRD paper typically begins with finding rules and unusual patterns in review data and culminates in (heuristic) model design. For example, Jindal et al. (2010) develop measures of *unexpectedness* in review behavior: confidence unexpectedness, support unexpectedness, attribute distribution unexpectedness and attribute unexpectedness. An example of a behavior called unexpected is that of a reviewer writing only positive or only negative reviews, or writing too many negative reviews, or too many positive reviews, etc. However, according to Wang et al. (2011), this method does not identify real spammers well. Sharma and Lin (2013) devise five more complex heuristic rules based on review content and ratings on Amazon.com. Peng and Zhong (2014) combine discriminatory rules with time series analysis.

Weighting Algorithms: A simple technical approach to FRD is to identify spam based on *spam scores*. Functions are defined to assess object spam scores from measurable indicators. A model is then built to combine multiple scores using weights to regulate

the contribution of all the scores into one classifier. The weights can be empirically set, or learnt from training data. The first weighting-based algorithm, due to Lim et al. (2010), integrates four spamming behavior models into spam scores for review writers. Lin et al. (2014) employ an unsupervised model to fine-tune weights in a calculated way. Shehnepoor et al. (2017) leverage a weighting algorithm to determine the importance of a spam feature in their graph-based framework, NetSpam. Allahbakhsh et al. (2013) use a weighting algorithm to detect collaboration between spammers.

6 Models exploiting network footprints (FRD-NF)

Observable behavioral and textual features alone may not be sufficient for FRD, because a spammer can write unbiased reviews for some products amid intentionally biased ones for other products (Wang et al. 2011). However, it is natural to assume that certain relations/dependencies may exist between the patterns found in review content, reviewer behavior and product/service characteristics, which may shed useful insights. For example, a spammer is likely to develop their own writing style while being in the business for a long time: this implies a review-reviewer relation. Spammers may be hired to write fake reviews for multiple products under the same brand: this feeds into reviewer-product relations. Some spammers may interact with each other in groups and use similar templates for fake review writing: this could explain reviewer-reviewer relations. Assuming that people (readers) are more likely to value the reviews from the peers they are connected with, it also make sense for spammers to be socially active to be effective (Xue et al. 2015). All such relations would be difficult to identify if the conventional feature-based methods described above were used in isolation of each other. Hence, much research has been devoted to developing FRD models that exploit the so-called network footprints.

In 2011, Wang et al. (2011) were first to argue that the truthfulness of a review, trustworthiness of a reviewer and reputation of a reviewed business, meaning the overall quality of their products, are interrelated and can be analyzed collectively, if represented by an attributed graph. Kumar et al. (2018b) introduce quality scores for users, ratings and products, embedded into a bipartite network model, and formulate the axioms that establish the inter-dependency between these scores, e.g.: (1) better products get higher ratings; (2) better products get more reliable positive ratings; (3) reliable ratings are closer to goodness scores; (4) reliable ratings are given by fairer users; and (5) fairer users give more reliable ratings. Lu et al. (2013) build an undirected review graph that includes all three actor types: reviews, reviewers and products. Wu et al. (2016) use a bipartite network of users and products where weights on edges represent ratings. Fei et al. (2013) study the interrelationships among reviewers only. Akoglu et al. (2013) define an FRD problem as a classification task on a bipartite network of reviewers and products, with signed edges representing review sentiment polarity. Rayana and Akoglu (2015) model relational data as an unsigned user—review—product network: they represent reviews as linked nodes, and integrate this information with the text, timestamps, and ratings to be able to use these meta-data for the attributes not only of users and products, but also of reviews. Based on Yelp

data, Wang et al. (2016a) define 11 types of interactive relations between reviewers and products in terms of time, locations, social contact, rating information, etc.

Overall, FRD-NF models are complex and mature, however, they are computationally expensive. Also, not all the necessary input information required for FRD-NF might be available for analysis in practice, due to data privacy concerns.

6.1 Graph-based modeling techniques

Iterative Computational Framework: Graph-based representations by Wang et al. (2011, 2012) have three types of nodes (review, reviewer, and business), with pre-defined node functions used for computing the values for such attributes as honesty of reviews, *trustiness* of reviewers and reliability of businesses. After inputting the available information into a review graph, an iterative computation framework (ICF) balances all observations, i.e., updates spam labels iteratively until convergence is achieved. Wahyuni and Djunaidy (2016) adopt the same model and integrate semantic aspects into it. Lu et al. (2013) design a *factor graph* to discover both fake reviews and spammers at the same time. Xue et al. (2015) represent relations among products and users in a social graph, weighted by *social proximity* between users. Li et al. (2019) develop a weighted reviewer-product bipartite graph informed by Yelp review data. Shah et al. (2017) perform FRD in a graph-based framework, EdgeCentric, which defines multi-role nodes, and relations between them, and focuses on evaluating *edge* attributes and *edge* patterns to produce a unified abnormality scoring function. Kumar et al. (2018b) offer an iterative algorithm, Rev2, for bipartite network analysis incorporating behavioral clues.

Probabilistic Graphical Modeling: Probabilistic Graphical Modeling (PGM) is a framework for modeling conditional dependencies between random variables (Koller and Friedman 2009). Directed graphical models—Bayesian Networks,—and undirected graphical models—i.e., Markov Random Fields (MRFs)—fall under PGM (Farasat et al. 2015). Hidden Markov Models, a special class of Dynamic Bayesian Networks (Farasat et al. 2015), are particularly well adapted for spammer detection (Li et al. 2016), while Markov Random Fields (MRFs) have become popular for the detection of both fake reviews and spammers (Fei et al. 2013; Akoglu et al. 2013; Li et al. 2014a; Rayana and Akoglu 2015; Xu et al. 2013; Fayazi et al. 2015). Fei et al. (2013) use MRFs to model relationships between reviewers and review co-occurrence (bursts). To uncover collusive spammers, Xu et al. (2013) offer a *k*-nearest neighbors method, and a PGM-based classification method on a pairwise MRF. Li et al. (2014a) use Dianping review data to build an undirected heterogeneous network of reviews, users and IP addresses, while capturing dependencies among the nodes. Akoglu et al. (2013) present an unsupervised network-based framework, FRAUDEAGLE, in which a MRF is trained to spot spammers and fake reviews in online review networks. Extending these ideas, Rayana and Akoglu (2015) propose another method, SpEagle, that utilizes relational data (review network) and meta-data (text, timestamps, ratings) collectively to detect spammers, fake reviews and targeted products. Fayazi et al. (2015) construct a reviewer-reviewer graph, and run an EM algorithm on an MRF for clustering, toward identifying groups of similar reviewers.

7 Models exploiting rating footprints (FRD-RF)

The approaches reviewed so far did not pay special attention to dissecting rating distributions for detecting spamming. FRD-RF models, which emerged around year 2010, do this, and have merit because: (1) it turns out possible to discover certain typical distributional patterns in the ratings submitted by truthful reviews, enabling anomaly detection; (2) ratings are easy to submit and analyze—they are readily available in the majority of datasets, independently of the presence or lack of accompanying review texts; and (3) working with numbers only (as in numeric ratings versus text) requires less computational power.

Feng et al. (2012b) posit that natural rating distributions, i.e., distributions that repeat/resemble each other, tend to emerge from truthful reviews, and that fake reviews distort these distributions. They hypothesize that there is a set of representative distributions for each given domain or platform: e.g., the shape of a rating distribution for a typical product on Amazon.com is asymmetric bi-modal (J-shaped)—with most high, some low and barely any neutral ratings (Hu et al. 2009). Hu et al. (2009) identify that users' bias is the reason behind the dominance of J-shaped distributions in product reviews. People only write reviews about a business/product if they are either extremely satisfied or dissatisfied after using/purchasing a service/product, which is called *purchasing* bias. Moreover, people do not bother writing a review if the service/product is average or "as expected", which is known as *under-reporting* bias, and this is another reason behind the lack of neutral ratings. With this in mind, Hu et al. (2009) recommend leveraging such features as the mode and standard deviation of rating distributions, as well as product price. In an initial FRD-RF work, in 2010, Wu et al. (2010b) confirm that fake reviews distort distributions of ratings for hotels (over multiple contributors), and establish a distortion measure as a signal, i.e., a validating criterion, for FRD. They define two scores that help quantify the magnitude of the distortion: *proportion of positive singletons* and *concentration of positive singletons*.

Many FRD-RF works only consider metadata (e.g., the number of reviews, average rating scores) as distinctive spam features (Rayana and Akoglu 2015; Martens and Maalej 2019; You et al. 2018; Liu and Pang 2018); they do this for two reasons: (1) such metadata are easily obtainable and computationally inexpensive to analyze; and (2) metadata-based models are domain-independent and generalizable, i.e., they can be easily implemented on any platform. Analysis of rating distributions provides a new perspective on spamming, but by itself, cannot inform anti-spamming policy-making. Review content offers richer evidence to support conclusions: indeed, expressed sentiment can justify the existence of generally positive reviews accompanied by low ratings, as well as of generally negative reviews with high ratings (Peng and Zhong 2014). Note also that two reviews giving the same rating to a product may have different influence on the reader due to the difference in review content.

The temporal aspects of rating footprints are also recognized and analyzed. As it was noted, fake review activity tends to take place in bursts (Fei et al. 2013); researchers also find that such bursts are temporally correlated with fluctuations in the average product/service rating. Indeed, if a large number of fake reviews target a product in a coordinated way, then the rating of the product abruptly changes, which was the point of the attack in the first place (Xie et al. 2012). With this in mind, researchers began

to use time-dependent spam indicators such as the *number of reviews per unit time*, *periodic ratio of singleton reviews*, *periodic ratio of positive reviews*, and *periodic average rating* for a given product/service/business.

In a 2012 paper, Xie et al. (2012) track several temporal correlation indicators for detecting singleton fake review attacks. Ott et al. (2012) hypothesize that deception attempts on a review platform depend on the cost (resource requirement for posting a review) and gain (average profit availed from a review) associated with producing the fake review. They explore the proliferation of deception over time in six popular online review communities: Expedia, Hotels.com, Orbitz, Priceline, TripAdvisor, and Yelp, and conclude that: (1) review platforms with “low cost and high gain” (e.g., TripAdvisor and Yelp) experience higher rates of fake reviewing; and (2) increasing the cost of fake-review-creation or introducing intervention policies is indeed helpful. Li et al. (2016) discover that both spammers and non-spammers exhibit bi-modal temporal posting (review submission) patterns, and these patterns significantly differ from each other. Ye et al. (2016) describe eight types of signals indicative of spamming, and extract time series data for individual products *en route* to performing FRD in real time, e.g., they track *youth score* (the average reviewer age at the time they reviewed a product) and *temporal gap entropy* (the time-gap between consecutive reviews). Analyzing restaurant reviews on Yelp.com, Santosh and Mukherjee (2016) discover temporal patterns of spam injection policies applied by review spammers such as *buffered spamming* and *reduced spamming*. When a business sees an increase in the number of truthful negative ratings (recently submitted), or a reduction in the number of positive ratings, the business may employ spammers to inject positive fake reviews to retain the popularity of its products to a certain level (threshold popularity): this phenomenon is termed *buffered spamming*. Naturally, spam injection rates may drop when a business maintains a decent popularity: this is termed *reduced spamming*. Minding these phenomena, Dong et al. (2020) present an FRD model leveraging multiple categories of reviewer-behavioral and review-content-based features, including rating signals, time signals and product-specific information.

To recap, the use of temporal footprints for FRD has a significant practical appeal since they are difficult to hide, and the models that rely on them are domain-independent.

7.1 Distributional and temporal pattern mining techniques

Distributional Pattern Mining: Special empirical approaches that go beyond visual inspection exist for mining distributional footprints. Feng et al. (2012b) examine rating distribution shapes on TripAdvisor and Amazon platforms. Beutel et al. (2014) present a Bayesian approach to Collaborative Filtering that can take nontrivial bi-modal and skewed distributions, along with conventional uni-modal distributions, as input for inference of true ratings.

Temporal Pattern Mining: A number of works consider fake reviewing a temporal phenomenon and approach FRD by performing anomaly detection in multivariate time series. Spam indicators plotted against time may reveal anomalous patterns traceable to spamming activities (Xie et al. 2012; Peng and Zhong 2014; Ye et al. 2016; Santosh

and Mukherjee 2016; Heydari et al. 2016). Xie et al. (2012) were first to do FRD using time series analysis relying only on rating data. Peng and Zhong (2014) integrate—both the rating scores and *discriminative* rules for sentiment information—into a time series analysis for FRD. Ye et al. (2016) devise an online FRD model that can identify abnormal events in real-time, leveraging eight time-series-based spam indicative signals. Heydari et al. (2016) leverage time series analysis to describe time windows during which spamming attacks are most likely to take place. Santosh and Mukherjee (2016) use a clustering algorithm for time series to discover temporal patterns using filtered (fake only) Yelp review data. Hooi et al. (2016) develop a Bayesian model for rating pattern analysis, BIRD, with an aim to combine both the temporal and rating distribution-based approaches.

8 Models exploiting collusion footprints (FRD-CF)

Spam attacks imply beneficiaries: it is natural to assume that spammers target specific products, product groups or services (Lim et al. 2010; Wang et al. 2011)—those which they are hired to promote. Such products should tend to be lower priced and have lower quality (Wang et al. 2011), thereby requiring artificial sales boosts. Further, for products that are not in demand, one can assume that a fake review should be among the first few reviews for such a product, or even end up being the only review for it. Thus, it makes sense to investigate target product(s), or product(s) likely to be targeted, and leverage product-centric features for FRD. In a 2008 paper, Jindal and Liu (2008) “evaluate” products by defining *sales rank* via price and average rating of a product. Li et al. (2011) leverage *product description* traits as features that may correlate with the number of fake reviews for a particular product.

One can also assume that if a particular product is targeted in an organized way, then the spammers are likely to be selected/hired as a group, perhaps from the same spam farm or through the use of referrals. Thus, another way to identify deceptive online reviews is to backtrack the “home base” of the spammers. Even though many communication channels outside a review platform can be used to coordinate spammers’ activities—including private methods such as email and instant messenger services, which are difficult to track,—the most notorious channel for hiring spammers is crowd-sourcing platforms. Such “dark”, i.e., illegally and anonymously maintained, platforms publicly advertise fake reviewer jobs. Acknowledging this, Fayazi et al. (2015) offer a sampling method for identifying products targeted for manipulation by crawling open jobs in a number of spam service marketplaces.

Note that in the entire FRD domain, only a few initial works have looked at product-based features. A reason for that may be the lack of available information specific to a product/service, and hence, the inability of researchers to describe vulnerable products as a group. Another possible reason is that product-based features do not directly help identify spam—only help reveal target products/services for which spamming activity might occur. Still, this is certainly valuable, since the possibility of successful identification of spam content opens a door to supervised learning, i.e., allows researchers to obtain subsets of data/reviews which can collectively help one define distinguishing features of spam. Collusive spamming have intrigued many researchers. One specu-

lation about spammers is that they may support each other's posts with up-votes and text replies. Organizing such collaboration, they develop communities/groups (Choo et al. 2015). A group of spammer accounts (either maintained by a single real person, or multiple persons, or combination of both) are believed to work together over a limited period of time to inject false content targeting a particular product. There are also other hypotheses: that spammer groups copy content of one another, and that they tend to strike right after a product becomes available for reviewing, to control the sentiment on it from the very beginning (Mukherjee et al. 2011). Independently of specific assumptions, the models that pay special attention to collusion deserve a separate discussion.

In the seminal work on spammer group identification, Mukherjee et al. (2011) introduce such special behavioral features as *time window*, *group deviation*, *group content similarity*, *early time frame*, and *group size*. Allahbakhsh et al. (2013) use rating score-based group features in their graph-based data-driven model to detect collusion. Xu and Zhang (2016) develop measures of homogeneity in collective behavior based on similarity in workload, timing of review posting, ratings given, etc. Wang et al. (2016b) propose such group spam indicators as review-, neighbor- and product-*tightness* (tightness is defined as the degree/extent to which review activities are similar among reviewers). They also describe interactions between spammer groups with such actions as “grow”, “break”, “merge”, etc. Arguing that identification of group behavior must be supported by similarity in individual behavior, Mukherjee et al. (2012) introduce additional features to detect potential members of the same spammer group: *individual rating deviation*, *individual early time-frame*, and *individual member coupling in a group*, among others. Figure 8 shows several reviews of a spammer group of three reviewers. Taken individually, these reviews do not appear suspicious, but several patterns become apparent in the group view: these reviewers have reviewed only these three products, gave the same ratings, and posted the reviews within a short time window following the product launch. Fei et al. (2013) work under the assumption that if a reviewer posts most of their reviews during the time period(s) when bursts of reviews appear, which may signal co-reviewing by the members of the same spammer group, then the reviewer is more likely to be a spammer. The caveat here is that reviewer submission during a burst does not necessarily make all such reviewers spammers; consequently, Li et al. (2016) propose to track co-bursting: instances of two or more occurrences where a pair of users post reviews during the rating burst(s) for the same product(s) (recall Fig. 6). Xu et al. (2013) include linguistic features into their model of collusion. Choo et al. (2015) and Wang et al. (2018) build user-relationship graphs that capture co-reviewing patterns, and take relationship strength as a spam signal.

Another proposition relevant to collusion detection is that physical locations of the spammers of the same group may be distributed (spatially) differently, compared to the population of benign users. Li et al. (2015) argue that long-term spammers may use multiple IP-addresses to register many more different accounts over a short period of time, and leverage temporal and spatial metrics, e.g., *average travel speed*. However, such an approach is effective for identifying fake reviews posted only for local businesses such as hotels and restaurants.

Profile of Big John	1 of 1 people found the following review helpful: ★★★★★ Practically FREE music, December 4, 2004 This review is from: Audio Xtract (CD-ROM) I can't believe for \$10 (after rebate) I got a program that gets me free unlimited music. I was hoping it did half what was
	3 of 8 people found the following review helpful: ★★★★★ Yes – it really works, December 4, 2004 This review is from: Audio Xtract Pro (CD-ROM) See my review for Audio Xtract - this PRO is even better. This is the solution I've been looking for. After buying iTunes,
	5 of 5 people found the following review helpful: ★★★★★ My kids love it, December 4, 2004 This review is from: Pond Aquarium 3D Deluxe Edition This was a bargain at \$20 - better than the other ones that have no above water scenes. My kids get a kick out of the
Profile of Cletus	2 of 2 people found the following review helpful: ★★★★★ Like a tape recorder...., December 8, 2004 This review is from: Audio Xtract (CD-ROM) This software really rocks. I can set the program to record music all day long and just let it go. I come home and my
	3 of 10 people found the following review helpful: ★★★★★ This is even better than...., December 8, 2004 This review is from: Audio Xtract Pro (CD-ROM) Let me tell you, this has to be one of the coolest products ever on the market. Record 8 internet radio stations at once,
	5 of 5 people found the following review helpful: ★★★★★ For the price you...., December 8, 2004 This review is from: Pond Aquarium 3D Deluxe Edition This is one of the coolest screensavers I have ever seen, the fish move realistically, the environments look real, and the
Profile of Jake	★★★★★ Wow, internet music!, December 4, 2004 This review is from: Audio Xtract (CD-ROM) I looked forever for a way to record internet music. My way took a long time and many steps (frustrating). Then I found Audio Xtract. With more than 3,000 songs downloaded in ...
	2 of 9 people found the following review helpful: ★★★★★ Best music just got, December 4, 2004 This review is from: Audio Xtract Pro (CD-ROM) The other day I upgraded to this TOP NOTCH product. Everyone who loves music needs to get it from Internet
	3 of 3 people found the following review helpful: ★★★★★ Cool, looks great...., December 4, 2004 This review is from: Pond Aquarium 3D Deluxe Edition We have this set up on the PC at home and it looks GREAT. The fish and the scenes are really neat. Friends and family

Fig. 8 Example of the activity of a spammer group (Mukherjee et al. 2012)

8.1 Behavioral pattern mining: rank aggregation techniques

Mukherjee et al. (2012) argue that feature-based FRD models—which have been successful against individual spammers—are less effective against *spammer groups*, and employ a rank aggregation method in combination with *frequent item-set mining*, to

uncover *groups of spammers*. One big disadvantage of rank-based methods, however, is that they do not directly identify fake reviews, and require the use of supervised classifiers on top of them.

Supervised Ranking: Learning to Rank is a common term for supervised rank aggregation algorithms, widely applied for *Information Retrieval*. With the goal of identifying spammer groups, Mukherjee et al. (2011) apply SVMRank, a learning-to-rank approach, to rank candidate spammer groups based on “spam indicators”. Mukherjee et al. (2012, 2013a) try other learning-to-rank techniques—SVMRank, RankBoost, etc.,—toward the FRD same purpose.

Unsupervised Ranking: Unlike *Learning to Rank*, unsupervised rank aggregation algorithms do not require large amounts of labeled data. However, these techniques require manually defined features/indicators. Typical features leveraged for rank aggregation models include rating information (Wu et al. 2010a), behavioral features (Mukherjee et al. 2013a), and relationship information (Mukherjee et al. 2012).

9 Implementation-related tools

This section discusses additional tools and techniques, developed to overcome implementation-related challenges for the approaches reviewed above.

Dataset Development: The need for large expert-labeled training data is the prime challenge for supervised learning for FRD. Banerjee (2019) categorizes the methods that can be adopted to obtain corpora of both truthful and fake reviews; the relevant methods that they discuss include manual annotation (Li et al. 2011), heuristic annotation (Jindal and Liu 2008), crowd-sourcing (Ott et al. 2011, 2013; Li et al. 2014c) and automation (Sun et al. 2013; Yao et al. 2017). Jindal and Liu (2008) develop training dataset by applying simple rules of thumb (heuristic annotation) e.g., reviews with similar textual content or irrelevant content are considered as “fake”, whereas reviews with unique, relevant content are considered “authentic”. They observe that identifying reviews as “fake” by simply reading them is extremely difficult; thus, they manually label only near-duplicate fake reviews and non-reviews. With the help of trained annotators, Li et al. (2011) manually build a set of fake reviews with the dataset extracted from Epinions.com, where users can evaluate posted reviews by providing comments and voting reviews as “helpful”. Li et al. (2011) trust the “wisdom of the crowd” and assume that most fake reviews must be “un-helpful” (per the Epinions.com user judgment). Note, however, that this logic works both ways, i.e., spammers may vote each other’s reviews as “helpful” as well. Few works attempt to artificially generate fake reviews e.g., Sun et al. (2013) convert an authentic review into a fake review by replacing the sentences of the authentic one with suitable similar content (selected from a corpus of review-text) using a semantic language model, whereas Yao et al. (2017) generate an initial fake review leveraging RNN-based generator and further customize it by replacing all the nouns with contextually-appropriate keywords. Mukherjee et al. (2013c) build a training dataset by using the commercial filtering algorithm of Yelp.com. The algorithm filters reviews based on different features including reviewer-behavioral features. In the resulting dataset, the filtered

Fake Review

“ My husband and I stayed for two nights at the Hilton Chicago, and enjoyed every minute of it! The bedrooms are immaculate, and the linens are very soft. We also appreciated the free WiFi, as we could stay in touch with friends while staying in Chicago. The bathroom was quite spacious, and I loved the smell of the shampoo they provided-not like most hotel shampoos. Their service was amazing, and we absolutely loved the beautiful indoor pool. I would recommend staying here to anyone. ”

Truthful Review

“ We stay at Hilton for 4 nights last March. It was a pleasant stay. We got a large room with 2 double beds and 2 bathrooms. The TV was Ok, a 27" CRT Flat Screen. The concierge was very friendly when we need. The room was very cleaned when we arrived, we ordered some pizzas from room service and the pizza was ok also. The main Hall is beautiful. The breakfast is charged, 20 dollars, kinda expensive. The internet access (WiFi) is charged, 13 dollars/day. Pros: Low rate price, huge rooms, close to attractions at Loop, close to metro station. Cons: Expensive breakfast, Internet access charged. Tip: When leaving the building, always use the Michigan Ave exit. It is a great view. ”

Fig. 9 One fake and one truthful hotel review [the dataset of Ott et al. (2011)]

reviews are taken as fake and unfiltered reviews as truthful ones. However, none of the above-mentioned datasets has been widely recognized as “reliable”.

In 2011, Ott et al. (2011) were first to compose the first gold-standard dataset: it consists of positive fake reviews (created by crowd-workers at Amazon Mechanical Turk) and positive truthful reviews (collected from TripAdvisor.com). Ott et al. (2011) manually label training examples for truthful opinions from reviews of 20 most popular Chicago hotels from TripAdvisor, based on the assumption that popular offerings are less likely to become target of fake reviews. They select 400 truthful reviews based on different exclusion criteria. For instance, they discard reviews which are not English, not 5-star, of less than 150 characters in length or written by first-time authors. Many works have relied on this dataset for model training (Xu and Zhao 2012; Li et al. 2013; Ren et al. 2014; Li et al. 2014c; Sandulescu and Ester 2015; Hai et al. 2016; Rout et al. 2017a). Figure 9 gives example of two positive reviews (compliments) of a hotel from the dataset due to Ott et al. (2011)—one fake review and one is a truthful review.

In 2013, Ott et al. (2013) incorporated negative sentiment reviews into their original dataset (Ott et al. 2011), by gathering negative truthful reviews from travel websites Expedia, Hotels.com, Orbitz, Priceline, TripAdvisor, and Yelp. Li et al. (2014c) offer a dataset comprised of reviews from three domains—hotels, restaurants, and medical doctors—including truthful reviews from customers and fake reviews from crowd-workers and domain-experts. Another dataset, due to Banerjee et al. (2015), is a corpus of 900 truthful reviews (collected from Agoda.com, Expedia.com and Hotels.com) and 900 fake reviews (collected from 284 participants) for 15 popular hotels in Asia. More recently, Martens and Maalej (2019) released a dataset of fake-and official-reviews of mobile apps by collecting reviews from the Apple App Store. Note, however, that models trained on publicly available data may not be effective against spamming models that used the same data for training. Plotkina et al. (2020) compile a corpus of 1041 reviews of a restaurant service, comprising both positive and negative reviews produced via crowd-sourcing (using the workforce of Amazon Mechanical Turk (AMT)). They also considered the impact of different stimuli on the fake-review-writers and asked the writers to produce fake reviews “as if from different

perspectives”. For example, some writers were asked to pretend as if they were the employee of the restaurant (for which the fake reviews were written), in line with what was termed “social stimuli”. Also, some writers were receiving higher compensation for writing “convincing” fake reviews (“monetary stimuli”), whereas others were just asked to write some fake reviews (“no stimuli”).

Overall, observe that on-demand fake reviews, generated by crowdworkers, may not be representative of real-life fake reviews by “professional spammers” (Mukherjee et al. 2013c). Again, no filtering algorithm or expert annotator can reliably label a review as fake/authentic, and even the well-accepted datasets have been manufactured one way or another.

One must be even more cautious building datasets by leveraging artificially-generated fake reviews, as it is unclear whether the distribution of the artificially-generated fake reviews resembles the distribution of the spammer-written fake reviews (which is unknown). Unbalanced data is another challenge for supervised classifiers: on any active platform, the number of fake reviews is much lower than that of truthful reviews (Mukherjee et al. 2013b; Luca and Zervas 2016), which complicates classifier training. To overcome this challenge, researchers perform *random under-sampling*: a balanced dataset is prepared by randomly selecting a subset of majority instances (truthful reviews) and combining them with all the minority instances for learning (Zhang et al. 2016; Fontanarava et al. 2017; Mukherjee et al. 2013c; Najada and Zhu 2014; Dong et al. 2018).

Identification of Spammer Groups: Frequent Item-set Mining (FIM) has been adopted to detect spammer groups as “frequent itemsets” (Li et al. 2016; Mukherjee et al. 2011, 2012; Allahbakhsh et al. 2013). In FIM, reviewers are viewed as items and reviewed products are viewed as transactions; the algorithm output is a “cohesive” user group. Truthful reviewers, however, may also be recognized as suspicious by this algorithm, and hence, additional measures are needed to fine-tune FIM (Mukherjee et al. 2012); e.g., Xu et al. (2013) do it by running a *maximal FIM* to identify only one, the largest group of likely spammers.

Wang et al. (2016b) identify several shortcomings in using FIM to generate candidate spammer-groups and propose a *loose* spammer group detection technique. Wang et al. (2018) model spammer groups as bi-connected instead of *loosely* connected. The statistical model of Xu and Zhang (2016) uses homogeneity-based collusive behavior measures. Li et al. (2016) argue that unlike co-bursting, co-reviewing does not necessarily indicate spamming, again stressing that co-reviewing-based approaches are computationally costly and ineffective with loosely connected subgraphs. They build a co-bursting network for FRD, by leveraging review time stamps.

Review Burst Detection: Fei et al. (2013) employ *Kernel Density Estimation* technique to detect spammers who write fake reviews in bursts. Xie et al. (2012) use a three-step approach for burst detection: it begins with curve fitting with a Bayesian change point detection algorithm on time series, followed by a template matching algorithm for identifying *bursty* pattern, and culminates in finding joint bursts in multi-dimensional time series. Heydari et al. (2016) also use time series to identify burst patterns that match certain templates.

Feature Selection Techniques: FRD-WF approaches typically use discrete textual features. Review texts are converted into word vectors, producing hundreds of thousands

of features, which become computationally expensive to process (Crawford et al. 2016). Managing and reducing feature subsets is thus an important task. Crawford et al. (2016) compare two techniques for reducing feature subset size in the FRD domain: filter-based and word frequency-based feature selection.

As manually-defined linguistic and psychological features may not be sufficient to capture global semantic information in text (see Sect. 3.1), Ren and Zhang (2016), Ren and Ji (2017), Li et al. (2017) and Wang et al. (2017) advocate for using Neural Networks (NNs) for modeling. In contrast to conventional feature-based approaches, NNs leverage a layered feature superimposition structure to automatically build complex features, capturing global semantic information from a sentence/discourse (Ren and Zhang 2016). NNs can be used to induce document-level representations from sentence-level information, which is difficult, if not impossible, to accomplish with conventional feature-based approaches and discrete hand-picked features. Ren and Zhang (2016) and Ren and Ji (2017) propose an FRD framework that leverages document representations. Zhang et al. (2018) train recurrent and convolutional NNs to learn to represent each word in reviews using contextual information and word embedding. They use Stochastic Gradient Descent to parameterize the NN. Doing benchmarking with cross-domain data, Ren and Ji (2017) conclude that NNs are more easily adaptable to new domains compared to discrete models. Wang et al. (2017) design a convolutional NN to jointly encode the textual and behavioral information toward review embedding. You et al. (2018) use NNs to jointly embed metadata and relational information for representation of review, reviewer, and product objects.

A Sentence-Weighted NN (SWNN) model, due to Li et al. (2017), incorporates importance weights for all individual sentences in review text. Li et al. (2019) develop a reviewer behavior representation model by integrating the entities and their social relations through a NN. Dong et al. (2020) use Autoencoder, a type of NN, to generate hidden representations of behavioral and textual features. They also use a stochastic and *differentiable* decision tree for model parameter learning. Wang et al. (2016a) focus on the challenge of learning representations of reviews directly from data and develop a model based on tensor factorization, for this purpose.

Note that while NN-assisted models are effective for extracting features, these models are not interpretable, i.e., do not allow one to intuitively understand and explain their logic (Dong et al. 2018).

Graph Clustering: In FRD-NF, clustering algorithms help reveal higher-level associations in review-based graphs—those among groups of nodes. Akoglu et al. (2013) employ the *Cross-Associations* clustering algorithm to identify spam attack groups. Mukherjee et al. (2013a) approach FRD as an unsupervised clustering problem in the Bayesian setting with two clusters. Ye and Akoglu (2015) design a clustering method to extract colluding groups of spammers from a sub-network of suspicious products and reviewers. Li et al. (2016) use hierarchical clustering along with two other clustering methods—Louvian and K-means. Santosh and Mukherjee (2016) offer the *K-spectral Centroid* clustering algorithm for time series.

Inference Techniques: The most diverse techniques for FRD are inference techniques. Inference is an optimization problem that lies in finding the labels of unclassified objects. Exact inference is known to be NP-hard for undirected graphical models such as MRFs; the number of possible class label combinations grows exponentially

with network size (Rayana and Akoglu 2015). Thus, approximate inference algorithms have been developed: in (social) graph-based works, approximate inference is typically based on belief propagation (Fei et al. 2013; Akoglu et al. 2013; Rayana and Akoglu 2015; Lu et al. 2013), Iterative Classification (Xu et al. 2013; Li et al. 2014a), Gibbs Sampling (Ott et al. 2012; Li et al. 2013; Mukherjee et al. 2013a; Xu et al. 2015; Mukherjee et al. 2016; Dong et al. 2018) or node ranking techniques (Shah et al. 2017; Kumar et al. 2018b).

Loopy belief propagation (LBP) is an iterative message-passing algorithm for running approximate inference for PGMs (Fei et al. 2013). LBP infers the posterior state probabilities for all network nodes, given the observed states of some of them. Another popular approximate inference technique, Gibbs sampling, works by sampling each variable, in turn, from the conditional distribution of that variable, given all other modeled variables. The conditional distribution of each variable can be derived from the joint distribution, read directly from the graph (Ott et al. 2012).

Addressing Cold-Start: The cold-start problem in model training in general, and in FRD in particular, stems from the fact that real-world data records for many *individual* objects are scarce, making it hard to make accurate inference about them. For example, when a new reviewer joins a platform, a behavioral feature-based classifier is helpless until it accumulates some review history. To mitigate this issue, Wang et al. (2017) cross-match behavioral patterns of similar reviewers. When a new reviewer posts a review, the textual information from that review is matched with pre-existing reviews, and multiple previously learnt behavioral patterns are fitted to the new reviewer. Similarly, You et al. (2018) leverage meta-data (locations, dates, purchase price ranges) of various entities (reviewers, reviews and businesses) to learn entity representations using deep learning. In their Rev2 model, Kumar et al. (2018b) add *Laplace smoothing* to the evaluation of fairness and goodness scores (to label reviewers and products, respectively), and build default prior beliefs for new entities. Li et al. (2019) also argue in support of behavior representation, as opposed to review representation, for mitigating cold-start.

The discussed *Implementation-Related Tools* are not only essential for model-building, but also beneficial for making FRD models more flexible and computationally tractable.

10 Known commercial implementations

This section reviews commercial tools, rooted in and relevant to the science of FRD, that are used at online e-commerce portals. Unfortunately, very few of such tools adopted by review-based platforms are publicized. For this reason, in this section, we cast a wider net and collect scientific papers about the practical use of the state-of-the-art works on “false information”, i.e., we review the papers on commercially implemented tools that combat opinion-based and fact-based false information dissemination. OSN service enterprises, as well as e-commerce and review aggregation platforms are the key stakeholders in the development of false-information detection techniques. Some of them, including Facebook, Tuenti, Flipkart and Renren, publish their works. Stein et al. (2011) report on the *Facebook Immune System*, a defense sys-

tem against fraud, phishing and spam, that performs real-time scanning of the contents on its continuously updated database. Boshmaf et al. (2011) report revealing Facebook Immune System's weaknesses through staged targeted attacks on its users by deploying a testing botnet. Beutel et al. (2013) propose CopyCatch to prevent group attacks on Facebook to defend against *bursty* spam activity: their algorithm searches for coordinated behavior and dense activity blocks in system logs. According to Cao et al. (2014), CopyCatch pioneered the work on aggregate behavior clustering in OSNs. However, CopyCatch cannot detect non-overlapping and partially overlapping synchronized behaviors and does not prevent repeated fraudulent activities from the same IP addresses (Cao et al. 2014).

According to Wang et al. (2013), "befriending normal users and then spamming them" is the strategy adopted by spammers that use Sybil (aka. pseudonymous) accounts. Analyzing click-stream data to detect malicious accounts in Renren social media website, Wang et al. develop three models—a click sequence model, time-based model, and hybrid model—to identify clusters of similar click patterns and consequently perform classification of likely Sybil behavior. However, this semi-supervised detection algorithm cannot handle compromised accounts well, since click-stream behavior of compromised accounts produces a mixed signal (Ruan et al. 2016). Another work based on Renren social network is due to Yang et al. (2011): they extract features from users' profiles and analyze link creation behavior of Sybil accounts by applying a SVM classifier, to find that Sybil accounts do not form tight-knit communities, which is contrary to the prior belief. Further, they report that a majority of connections among Sybil accounts are accidental, and also, that many authentic users befriend Sybil accounts voluntarily, which makes them even more difficult to identify with graph-based detection techniques.

Cao et al. (2012) introduce SybilRank, a random-walk-based Sybil detection scheme that traverses the social graph structure. It works under the assumption that authentic users have limited connectivity with Sybil ones. SybilRank was deployed on Tuenti, the largest OSN in Spain. However, SybilRank turned out not to be scalable for large networks, and further, suffered from targeted attacks aimed at gaming it, since its effectiveness largely depends on social network distance between Sybil account groups and well-connected clusters of trusted seeds.

Hooi et al. (2016) develop BIRDNEST algorithm to detect review spammers on Flipkart: its Bayesian model detects anomalies in rating distribution, considering temporal user activity information, and outputs a likelihood-based suspiciousness metric for users and reviews. Rev2 and Fairjudge algorithms, also developed to identify review spammers on Flipkart network, leverage both network-based and behavior-based properties to calculate quality metrics such as fairness of a user, reliability of a rating, and goodness of a product (Kumar et al. 2018b, 2017). Kumar et al. claim that, based on precision and AUC score, both these algorithms outperform the competitors, including BIRDNEST, while offering scalability and providing theoretical convergence guarantees. They also state that both Rev2 and Fairjudge were deployed on Flipkart to detect fake review writers. However, they did not consider temporal evolution of the behavior-based properties (Kjartansson and Shabbir 2017), and their algorithms required additional node-level information to differentiate spammer groups from other spammers. Soon thereafter, Kjartansson and Shabbir (2017) report on hav-

ing outperformed Fairjudge on AUC score, with an algorithm that employs a relatively fast time series analysis.

Review forums and e-commerce sites also use policy-making to curb fake review activity. E-commerce giant Amazon.com takes immediate actions against the owners of accounts suspected of manufacturing fake reviews to increase their sales. Amazon.com also displays *verified purchase* signs for some reviews speaking to their authenticity (Mukherjee et al. 2012), and allows users to identify reviews as *helpful* or *not helpful* as another means of validating review quality (Lim et al. 2010). However, spammers find ways to bypass such verification systems. For example, a group of spammers can *vote up* fake reviews written by spammer peers, claiming them *helpful* (Mukherjee et al. 2013a). To increase the credibility of some products, Amazon.com introduced *Amazon Vine* and *Vine Voice* programs to allow a select group of “Expert” reviewers to provide reviews about new and pre-release products. However, this service is too expensive for smaller businesses to afford, as only “enlisted” vendors/manufacturers (not the sellers) are able to promote their products this way. Amazon.com took another initiative, *Early Reviewer* program, to address the cold-start problem for newly launched products by acquiring early reviews. Under this program, a randomly selected group of early purchasers are asked to share their experience with the product, to help the fellow customers make informed decisions about it.

Spammers often use social media platforms such as Facebook.com⁴ and crowd-sourcing platforms such as Amazon Mechanical Turk (Fayazi et al. 2015) to produce malicious reviews by crowd-sourced workforce in exchange for free products and commission. Such a reviewer is asked to buy a product, thus being able to claim a *verified purchase*, and then receives the refund via hidden channels as soon as the platform posts the “verified” fake review⁴. Review forums such as Yelp.com introduced filters to prevent fake review activity. According to Mukherjee et al. (2013c), Yelp has a proprietary algorithm to identify fake reviews. Mukherjee et al. (2013c) tried to identify how the Yelp’s algorithm works, by analyzing its filtered reviews, and concluded that Yelp likely uses a behavioral pattern-based approach, leveraging features extracted from meta-data, including mouse gestures, click behaviors, and social network interactions among reviewers. In the same vein, Walmart.com claims that they moderate all reviews on their website.⁵

All the approaches mentioned above have made valuable contributions to the well-being of the respective online platforms. Of course, these approaches benefit from the full access to the data and system infrastructure, and can only be implemented by platform owners.

11 Concluding remarks and future research considerations

This systematic literature review showed how the FRD techniques progressively increased in complexity in response to the evolving spamming techniques, by sum-

⁴ 5-star phonies: Inside the fake Amazon review complex. <https://thehustle.co/amazon-fake-reviews>.

⁵ Fake reviews increasing among major retailers. https://www.cbs.com/shows/cbs_this_morning/video/3aql_Fh_pZ1x29R_uN7E4HOcsqEWQad/increase-in-fake-reviews-hitting-walmart-amazon-and-other-retailers/.

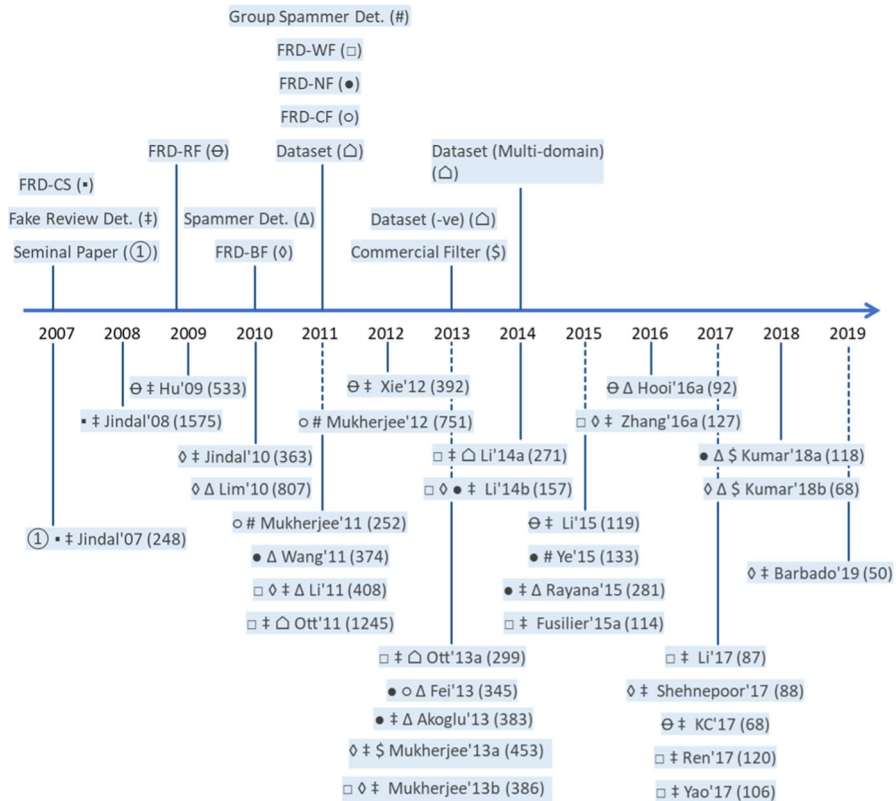


Fig. 10 Chronology of the FRD research approaches, in reference to the most influential articles (2007–2019)

marizing the contributions of the state-of-the-art FRD articles. It categorized FRD *modeling approaches* by the researchers' assumptions regarding how fake reviews get generated and how review spammers behave. Figure 10 shows the chronology of the FRD research by capturing the key milestones, along with referencing the most influential FRD articles, on the timeline between years 2007 and 2019. In the Fig. 10, the milestones are shown above the timeline, each labeled with a unique symbol. The cornerstone FRD articles are shown below the timeline, accompanied by the symbols that match them with the milestones, and also, the citation counts (in parenthesis) as of early 2021.

Summarizing the findings from Sects. 2–10, we see that no work to date has been able to once-and-for-all solve the FRD problem. While more methods are being developed addressing drawbacks of previously proposed methods, all the approaches reviewed in this paper have their limitations. The unresolved or unresolvable challenges can be summarized as follows:

1. professional attackers are well-equipped to mimic the behavior of truthful reviewers, making FRD methods (leveraging the respective textual and/or social features) less effective;
2. most FRD approaches are reactive and cannot perform in real-time; as a result, they often fail to prevent the damage caused by fake reviews and fraudsters; even if spam influence is eventually detected, sometimes it's too late to do any reputation-repairs;
3. feature-based approaches require training data in order to achieve accurate spam detection; construction of training datasets is time-consuming and costly; again, there is a valid concern that the FRD models, built in reliance on any given training dataset, are vulnerable to attacks that leverage information from the same dataset; additionally, availability of real user data in practice may be limited due to user privacy concerns;
4. recently developed algorithms that use textual, behavioral, temporal, rating-based and graph-based features in aggregation tend to be slow and not scalable;
5. most reported approaches are designed to cater for a particular platform, which makes their performance domain-specific and less flexible; and
6. all these approaches fail to distinguish between malicious reviews and truthful atypical reviews, which leads to unnecessary harassment of truthful reviewers.

All this said, there exists another issue that deserves special attention. The key conclusion from our literature review is that the current FRD methods are developed as counter-actions against continuously evolving spamming techniques, with spammers always being one step ahead.

Filtering fake reviews and removing them is a task that all large review platforms have attended to for a long time. Review forums and e-commerce sites such as Yelp, Amazon, Flipkart, etc., deploy proprietary FRD algorithms. More specifically, Flipkart quietly deletes fake reviews, Amazon publicly acknowledges review censoring, whereas Yelp keeps fake reviews visible to the public while flagging them as “potentially fraudulent” (Ananthakrishnan et al. 2020). These platforms do not reveal how their filtering algorithms work. Yelp argues that the opacity of their filtering algorithm is a design choice aimed to prevent system gaming. Also, transparency in an algorithm's operation might complicate users' interactions with the system. Note, however, the very fact that a platform can remove any content at-will raises concerns about corporate censorship (Lazer et al. 2018). Indeed, there is a growing mistrust among consumers toward technology platforms (Ananthakrishnan et al. 2020). “Conflicts of interest” is a major reason behind it: a conflict between a business and a review platform can occur if: (1) the business is both advertised and reviewed on the same platform; or (2) the business and the platform become competitors by selling the same products to consumers. For example, Yelp has been sued (unsuccessfully) by businesses for hiding positive reviews as “suspected fake” and keeping likely-fake negative reviews untouched as a way to extort businesses into advertising, and has been the subject of many related conspiracy theories regarding review censoring behavior.

Perhaps, the new-age solution against fake reviews should be decentralized. Indeed, it should be possible to promote reviewing among honest consumers. According to Lai and Tan (2019), humans should be the final decision makers in critical tasks that involve ethical and legal concerns such as combating fake reviews. Instead of taking control

of content filtering, platforms could enable users to navigate toward the content they value, and allow users to choose their own trusted voices. Two issues exist, however: (1) this scheme will not be effective if a user nominates only few peers as trustworthy; and (2) it will take time for a new user to embed themselves into a trusted network circle. It would be very difficult for consumers to build such individual trust networks that would give them sufficient feedback on a broad enough subset of the products and services, the modern platforms host. This demands new policy-making ideas which will be devoid of these issues while still involving users into trust-building.

According to the interdisciplinary work by Lazer et al. (2018), two initiatives can be undertaken to curb the influx of false information in online platforms: (1) user empowerment; and (2) information system structure revision that will actively prevent user *exposure* to false information. Citing the call for action per Lazer et al. (2018), and relying on the findings of this systematic literature review, we also recommend for the research community to move toward interventions and policies that would leverage social relationships among people toward formation of trustworthy communities on review platforms.

In conclusion, in the fight against fake reviews, there are promising prospects for trusted community-based information systems. Also, multiple research directions, both theoretical- and application-based, are open for exploration, which can eventually transform the idea of restructuring the information systems into a viable solution against fake review influence.

Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this article.

A Appendix

The FRD approaches reviewed in this paper are summarized in Tables 2 and 3. Table 2 organizes the modeling approaches. Table 3 covers the technical approaches for FRD. Table 1 recaps the used abbreviations and terms, for reader convenience.

Table 1 Definitions of the properties of Fake Review Detection approaches

Properties	Definition
FRD-CS	Approaches that detect fake reviews based on similarity in content by utilizing similarity measures
FRD-WF	Approaches that focus on writing footprints to classify reviews and utilize text-based features
FRD-BF	Approaches that analyze reviewing behaviors of reviewers to detect fake reviewers
FRD-NF	Approaches that identify fake reviews and spammers by leveraging the relationships among reviewers, reviews and products/business
FRD-CF	Approaches that search for collusion footprints to identify group of spammers
FRD-RF	Approaches that focus on rating footprints and utilize temporal and meta-data features to reveal fake reviews and spammers
Singleton	Approaches that target fake reviewers who write only one review
Group Spammer	Approaches that target group of spammers who work in a synchronized fashion
Spammer	Approaches that target individual fake reviewers
Fake Review	Approaches that focus on identifying fake reviews only

Table 2 Fake Review Detection papers organized by modeling approaches and datasets used

Paper	Year	Dataset	FRD-CS	FRD-WF	FRD-BF	FRD-NF	FRD-RF	FRD-CF	F. Review	Spammer	Group	Singleton
Jindal and Liu	2007a	Amazon	✓						✓			
Jindal and Liu	2007b	Amazon	✓						✓			
Jindal and Liu	2008	Amazon	✓						✓			
Hu et al.	2009	Amazon				✓			✓			
Algur et al.	2010	Epinion	✓						✓			
Jindal et al.	2010	Amazon		✓					✓			
Lai et al.	2010a	Amazon	✓						✓			
Lai et al.	2010b	Amazon	✓						✓			
Lim et al.	2010	Amazon		✓						✓		
Wu et al.	2010	TripAdvisor		✓					✓			✓
Lau et al.	2011	Amazon	✓						✓			
Li et al.	2011	Epinion		✓					✓	✓		
Mukherjee et al.	2011	Amazon						✓				
Ott et al.	2011	Developed own dataset		✓				✓	✓			
Wang et al.	2011	Resellerratings				✓				✓		
Wu et al.	2011	TripAdvisor					✓		✓			✓
Feng et al.	2012a	TripAdvisor & Amazon		✓			✓		✓			
Feng et al.	2012b	Ott et al. (2011)	✓					✓				
Harris	2012	Amazon & others		✓					✓			

Table 2 continued

Paper	Year	Dataset	FRD-CS	FRD-WF	FRD-BF	FRD-NF	FRD-RF	FRD-CF	F. Review	Spammer	Group	Singleton
Mukherjee et al.	2012	Amazon						✓			✓	
Ott et al.	2012	Yelp & 5 others					✓		✓			
Wang et al.	2012	Resellerratings		✓						✓		
Xie et al.	2012	Resellerratings				✓			✓			✓
Xu and Zhao	2012	Ott et al. (2011)							✓			
Akoglu et al.	2013	SWM	✓						✓			
Allahbakhsh et al.	2013	Amazon				✓		✓		✓	✓	
Fei et al.	2013	Amazon				✓		✓				
Feng and Hirst	2013	Ott et al. (2011)	✓						✓			
Fusilier et al.	2013	Ott et al. (2011)	✓						✓			
Li et al.	2013	Ott et al. (2011)	✓						✓			
Lu et al.	2013	Amazon		✓					✓	✓		
Mukherjee et al.	2013a	Amazon		✓							✓	
Mukherjee et al.	2013b	Yelp & AMT	✓	✓					✓			
Mukherjee et al.	2013	Yelp & AMT		✓					✓			
Ott et al.	2013	Developed own dataset	✓						✓			
Sharma and Lin	2013	Amazon	✓	✓					✓			
Sun et al.	2013	TripAdvisor	✓						✓			
Xu et al.	2013	Amazon.cn			✓			✓			✓	
Banerjee and Chua	2014a	Ott et al. (2013)	✓						✓			

Table 2 continued

Paper	Year	Dataset	FRD-CS	FRD-WF	FRD-BF	FRD-NF	FRD-RF	FRD-CF	F. Review	Spammer	Group	Singleton
Banerjee and Chua	2014b	Ott et al. (2011)	✓						✓			
Beutel et al.	2014	Amazon & others					✓			✓		
Choo et al.	2014	Amazon			✓			✓				✓
Li et al.	2014a	Dianping	✓						✓			
Li et al.	2014b	Dianping	✓		✓				✓			
Lin et al.	2014	Amazon	✓		✓				✓			
Najada and Zhu	2014	Ott et al. (2011, 2013)	✓					✓				
Peng and Zhong	2014	Resellerratings					✓		✓			
Shojaee et al.	2014	Yelp & 6 others	✓						✓			
Banerjee et al.	2015	Developed own dataset	✓						✓			
Fayazi et al.	2015	Amazon & others						✓		✓		
Fusilier et al.	2015a	Ott et al. (2011, 2013)	✓						✓			
Fusilier et al.	2015b	Ott et al. (2011, 2013)	✓						✓			
Li et al.	2015a	AMT & Ott et al. (2011)	✓						✓			
Li et al.	2015b	Dianping					✓		✓			
Rayana and Akoglu	2015	eBay							✓	✓		
Ren et al.	2015	Ott et al. (2011)	✓						✓			
Savage et al.	2015	Amazon			✓					✓		
Xu et al.	2015	Amazon & others	✓						✓			

Table 2 continued

Paper	Year	Dataset	FRD-CS	FRD-WF	FRD-BF	FRD-NF	FRD-RF	FRD-CF	F. Review	Spammer	Group	Singleton
Xue et al.	2015	Yelp				✓				✓		
Ye and Akoglu	2015	Amazon & iTunes				✓					✓	
Crawford et al.	2016	Li et al. (2014c)	✓						✓			
Hai et al.	2016	Ott et al. (2011)	✓						✓			
Heydari et al.	2016	Amazon					✓		✓			
Hooi et al.	2016	FlipKart & SWM					✓		✓			
Li et al.	2016	Dianping					✓			✓	✓	
Mukherjee et al.	2016	TripAdvisor & Yelp	✓		✓		✓		✓			
Ren and Zhang	2016	Li et al. (2014c)	✓						✓			
Sandulescu and Ester	2016	Yelp & others	✓									✓
Wahyuni and Djunaidy	2016	Amazon							✓			
Wang et al.	2016a	Amazon				✓		✓			✓	
Wang et al.	2016b	Yelp	✓						✓			
Wu et al.	2016	Amazon & others			✓					✓		
Xu and Zhang	2016	Amazon(.com, .cn)						✓			✓	
Ye et al.	2016	FlipKart & SWM							✓			✓
Zhang et al.	2016	Yelp	✓		✓				✓			
Banerjee and Chua	2017	TripAdvisor & 9 others	✓						✓			
Fontanarava et al.	2017	Yelp	✓		✓		✓		✓			✓
Santosh and Mukherjee	2017	Yelp					✓		✓			

Table 2 continued

Paper	Year	Dataset	FRD-CS	FRD-WF	FRD-BF	FRD-NF	FRD-RF	FRD-CF	F. Review	Spammer	Group	Singleton
Li et al.	2017	Li et al. (2014c)	✓						✓			
Ren and Ji	2017	Li et al. (2014c)	✓						✓			
Rout et al.	2017a	Yelp & 4 others	✓						✓			
Rout et al.	2017b	Ott et al. (2011)	✓						✓			
Shah et al.	2017	Amazon & others				✓				✓		
Shehnepoor et al.	2017	Amazon & Yelp		✓					✓			
Wang et al.	2017	Yelp	✓						✓			
Yao et al.	2017	Yelp	✓						✓			
Aghakhani et al.	2018	TripAdvisor	✓						✓			
Cardoso et al.	2018	TripAdvisor & Yelp	✓						✓			
Dong et al.	2018a	Yelp	✓						✓			
Dong et al.	2018b	Amazon	✓		✓					✓		
Kumar et al.	2018a	Amazon & others				✓				✓		
Kumar et al.	2018b	Yelp			✓					✓		
Liu and Pang	2018	Amazon	✓		✓							
Wang et al.	2018	Amazon, Yelp					✓		✓			
You et al.	2018	Yelp	✓		✓				✓		✓	
Zhang et al.	2018	Li et al. (2014c)	✓						✓			
Barbado et al.	2019	Yelp			✓				✓			
Li et al.	2019	Yelp			✓	✓				✓		
Martens and Maalej	2019	Apple App Store					✓		✓			

Table 3 Fake Review Detection papers organized by technical approaches and tools

Paper	Year	Cls-SV	Cls-SSV	LM	Topic	Graph	PGM	Weight	Empirical	Rules	Time	Rank	Similarity	Dataset	Cold	Feature	Burst	Grp-det	Clustering	Inference
Jindal and Liu	2007a	✓											✓							
Jindal and Liu	2007b	✓											✓							
Jindal and Liu	2008	✓											✓							
Hu et al.	2009																			
Algur et al.	2010									✓										
Jindal et al.	2010																			
Lai et al.	2010a	✓											✓							
Lai et al.	2010b			✓									✓							
Lim et al.	2010							✓												
Wu et al.	2010										✓									
Lau et al.	2011			✓									✓							
Li et al.	2011	✓											✓							
Mukherjee et al.	2011		✓									✓				✓				
Ott et al.	2011	✓												✓						
Wang et al.	2011					✓														
Wu et al.	2011																			
Feng et al.	2012a	✓							✓											
Feng et al.	2012b	✓							✓											
Harris	2012	✓		✓										✓						
Mukherjee et al.	2012	✓										✓	✓					✓		

Table 3 continued

Paper	Year	Cls-SV	Cls-SSV	LM	Topic	Graph	PGM	Weight	Empirical	Rules	Time	Rank	Similarity	Dataset	Cold	Feature	Burst	Grp-det	Clustering	Inference
Ott et al.	2012	✓									✓									✓
Wang et al.	2012					✓														
Xie et al.	2012										✓						✓			
Xu and Zhao	2012	✓																		
Akoglu et al.	2013	✓					✓										✓			✓
Allabakhsh et al.	2013					✓		✓					✓				✓			
Fei et al.	2013	✓					✓				✓					✓				✓
Feng and Hirst	2013	✓																		
Fusilier et al.	2013	✓		✓																
Li et al.	2013	✓			✓															
Lu et al.	2013	✓				✓							✓							✓
Mukherjee et al.	2013a											✓								✓
Mukherjee et al.	2013b	✓		✓									✓							
Mukherjee et al.	2013	✓		✓									✓							
Ott et al.	2013	✓												✓						
Sharma and Lin	2013									✓										
Sun et al.	2013	✓											✓							
Xu et al.	2013	✓					✓										✓			
Banerjee and Chua	2014a	✓																		✓
Banerjee and Chua	2014b	✓																		

Table 3 continued

Paper	Year	Cls-SV	Cls-SSV	LM	Topic	Graph	PGM	Weight	Empirical	Rules	Time	Rank	Similarity	Dataset	Cold	Feature	Burst	Grp-det	Clustering	Inference
Beutel et al.	2014								✓											
Choo et al.	2014					✓														
Li et al.	2014a	✓		✓									✓							
Li et al.	2014b	✓		✓			✓													✓
Lin et al.	2014	✓						✓					✓							
Najada and Zhu	2014	✓												✓						
Peng and Zhong	2014									✓										
Shojaee et al.	2014	✓																		
Banerjee et al.	2015	✓												✓						
Fayazi et al.	2015	✓					✓													
Fusilier et al.	2015a	✓																		
Fusilier et al.	2015b	✓		✓																
Li et al.	2015a	✓			✓								✓							
Li et al.	2015b	✓																		
Rayana and Akoglu	2015						✓						✓							✓
Ren et al.	2015	✓		✓																
Savage et al.	2015								✓											
Xu et al.	2015	✓		✓									✓							✓
Xue et al.	2015					✓														
Ye and Akoglu	2015												✓							✓

Table 3 continued

Paper	Year	Cls-SV	Cls-SSV	LM	Topic Graph	PGM	Weight	Empirical	Rules	Time Rank	Similarity	Dataset	Cold Feature	Burst	Grp-det	Clustering	Inference
Crawford et al.	2016	✓											✓				
Hai et al.	2016	✓	✓														
Heydari et al.	2016							✓						✓			
Hooi et al.	2016						✓										
Li et al.	2016					✓								✓			
Mukherjee et al.	2016	✓			✓										✓		
Ren and Zhang	2016	✓											✓				
Sandulescu and Ester	2016				✓					✓							
Wahyuni and Djunaidy	2016	✓			✓												
Wang et al.	2016a				✓					✓							
Wang et al.	2016b	✓											✓				
Wu et al.	2016	✓	✓		✓								✓				
Xu and Zhang	2016							✓									✓
Ye et al.	2016						✓										✓
Zhang et al.	2016	✓								✓							
Banerjee and Chua	2017							✓									
Fontanarava et al.	2017	✓											✓				
Santosh and Mukherjee	2017									✓							✓
Li et al.	2017	✓								✓							
Ren and Ji	2017												✓				

Table 3 continued

Paper	Year	Cls-SV	Cls-SSV	LM	Topic	Graph	PGM	Weight	Empirical	Rules	Time	Rank	Similarity	Dataset	Cold	Feature	Burst	Grp-det	Clustering	Inference
Rout et al.	2017a	✓																		
Rout et al.	2017b		✓																	
Shah et al.	2017					✓							✓							✓
Shenepoor et al.	2017	✓						✓												
Wang et al.	2017	✓												✓		✓				
Yao et al.	2017	✓		✓									✓							
Aghakhani et al.	2018	✓																		
Cardoso et al.	2018	✓																		
Dong et al.	2018a	✓			✓								✓				✓			
Dong et al.	2018b	✓														✓				✓
Kumar et al.	2018a	✓					✓							✓						✓
Kumar et al.	2018b	✓														✓				
Liu and Pang	2018												✓							
Wang et al.	2018					✓											✓			
You et al.	2018	✓												✓		✓				
Zhang et al.	2018															✓				
Barbado et al.	2019	✓												✓						
Li et al.	2019															✓				
Martens and Maalej	2019	✓					✓							✓						

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